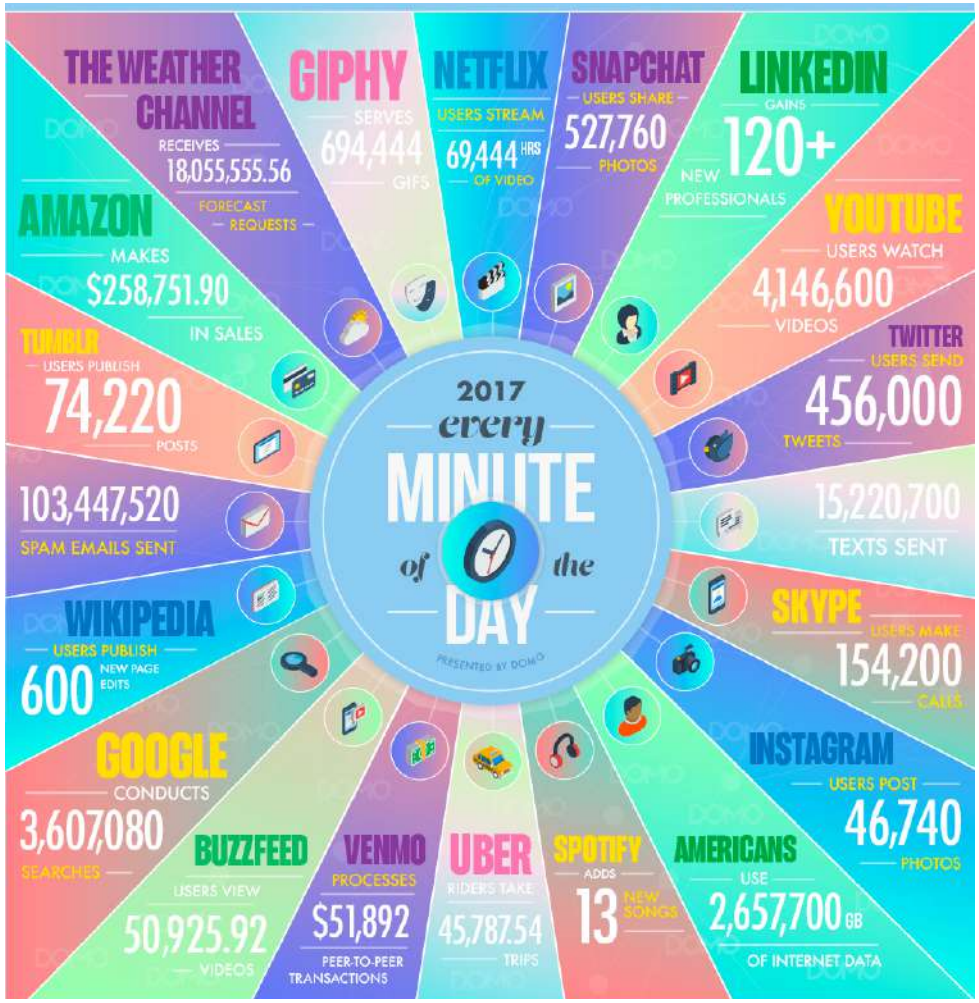
Toward Decision-driven Scheduling

Preliminary Exam

Aug. 20th of 2018

Presented by Jongdeog Lee

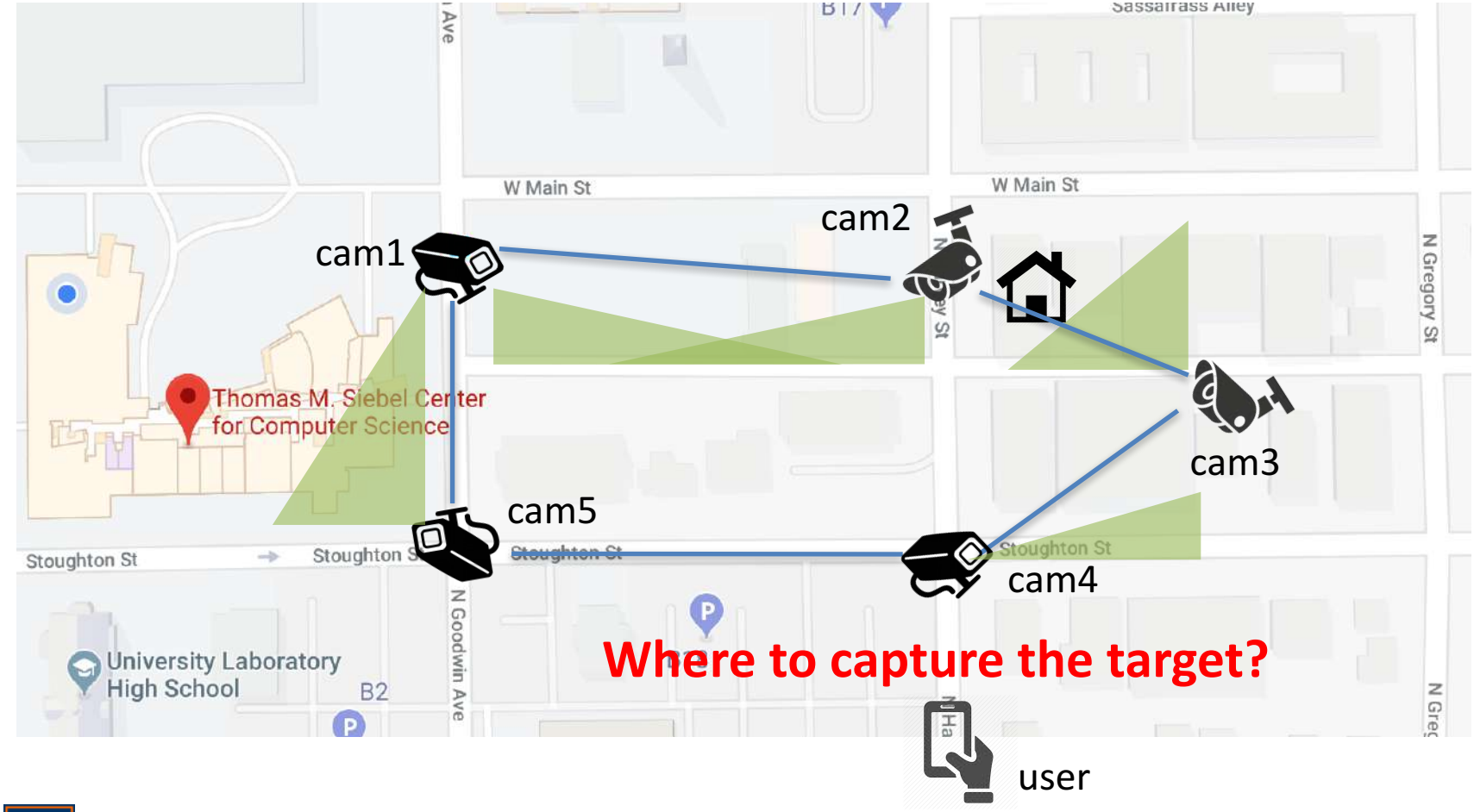
Every Minute …



Decision-driven Execution

- **User specifies decision to be made**
- **Network finds the best information to make the decision**
- **What vs. Why**
  - Network system can more efficiently find contents and sources to meet decision needs

Decision-driven Execution



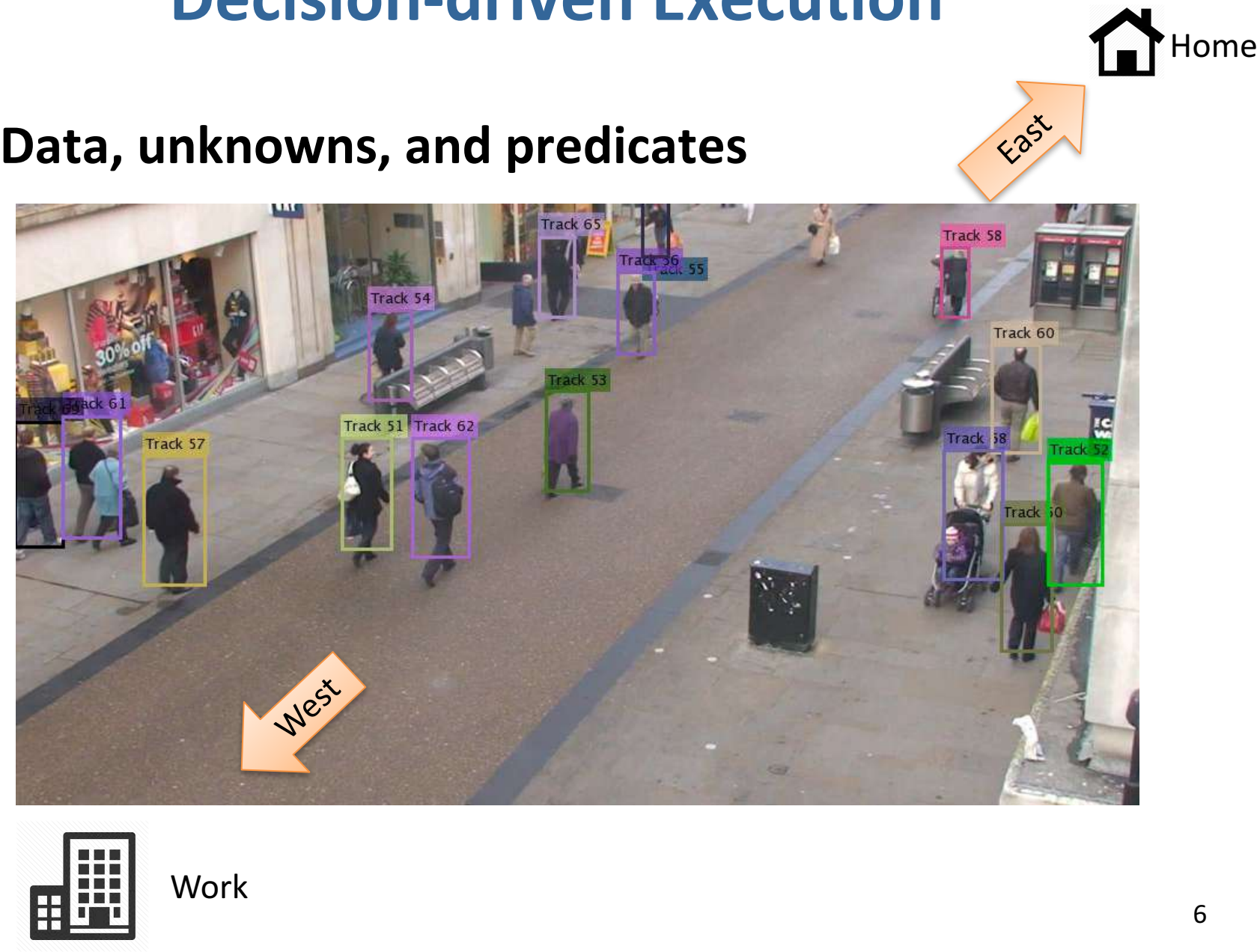
Decision-driven Execution

- **Abstraction**

Term	Description	Example
Decision	**A choice** among multiple courses of actions	Where to capture the target?
Courses of action	**Candidate answers** for the given decision	Home? At Work?
Validity logic	**Logical formula** to make the courses of action valid	Is the target heading home? Is the target heading to work?
Unknown	**The measurement** of physical environment	Current target location? Current target direction?
Predicate	**A logical condition** on the value of unknown	Current target location != Home? Current target direction == East?
Data	Data objects that **offer evidence** for determining the value of unknown	An image from camera 1 showing the target

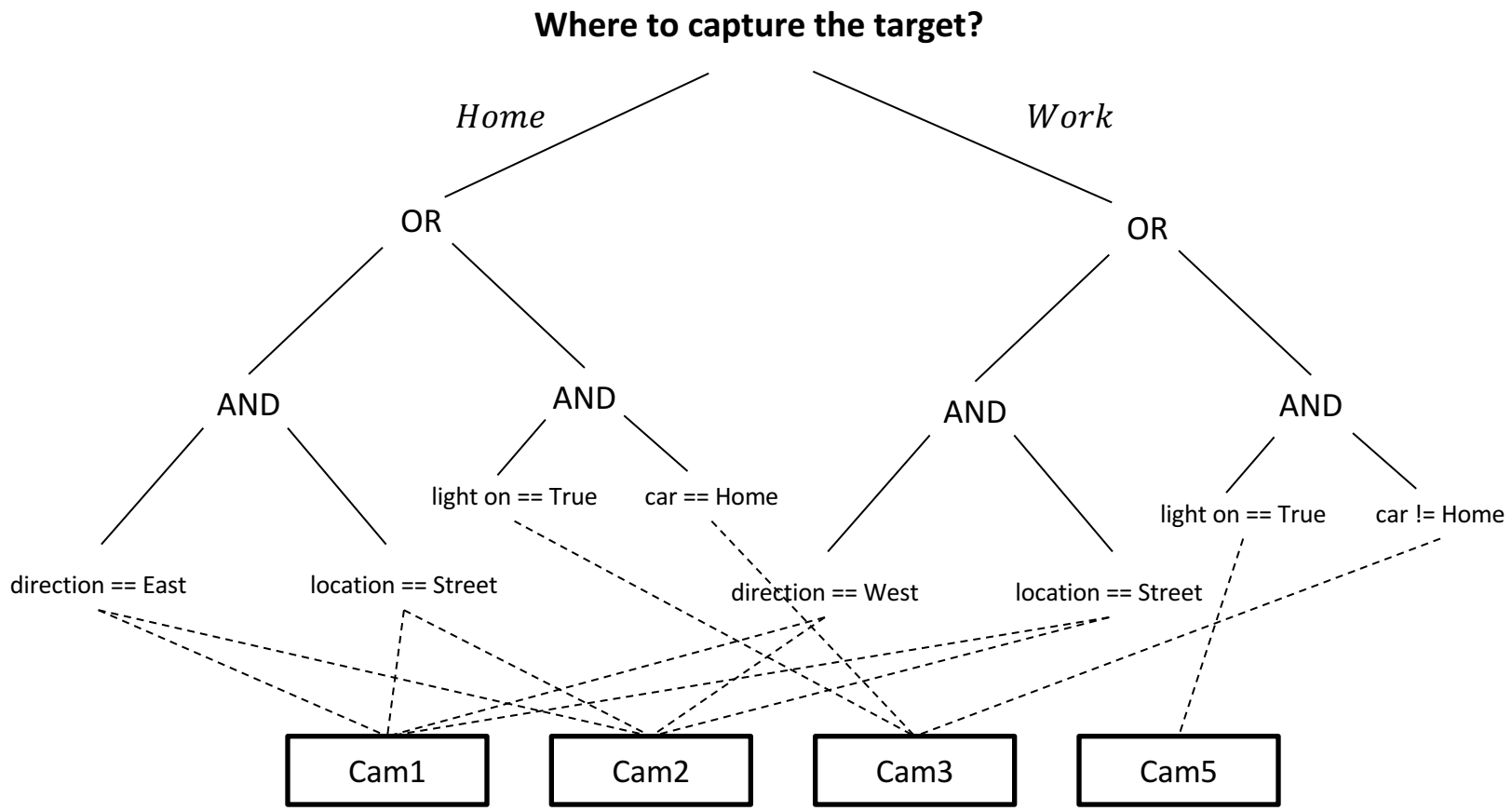
Decision-driven Execution

- **Data, unknowns, and predicates**



Decision-driven Execution

- **Decision Tree**

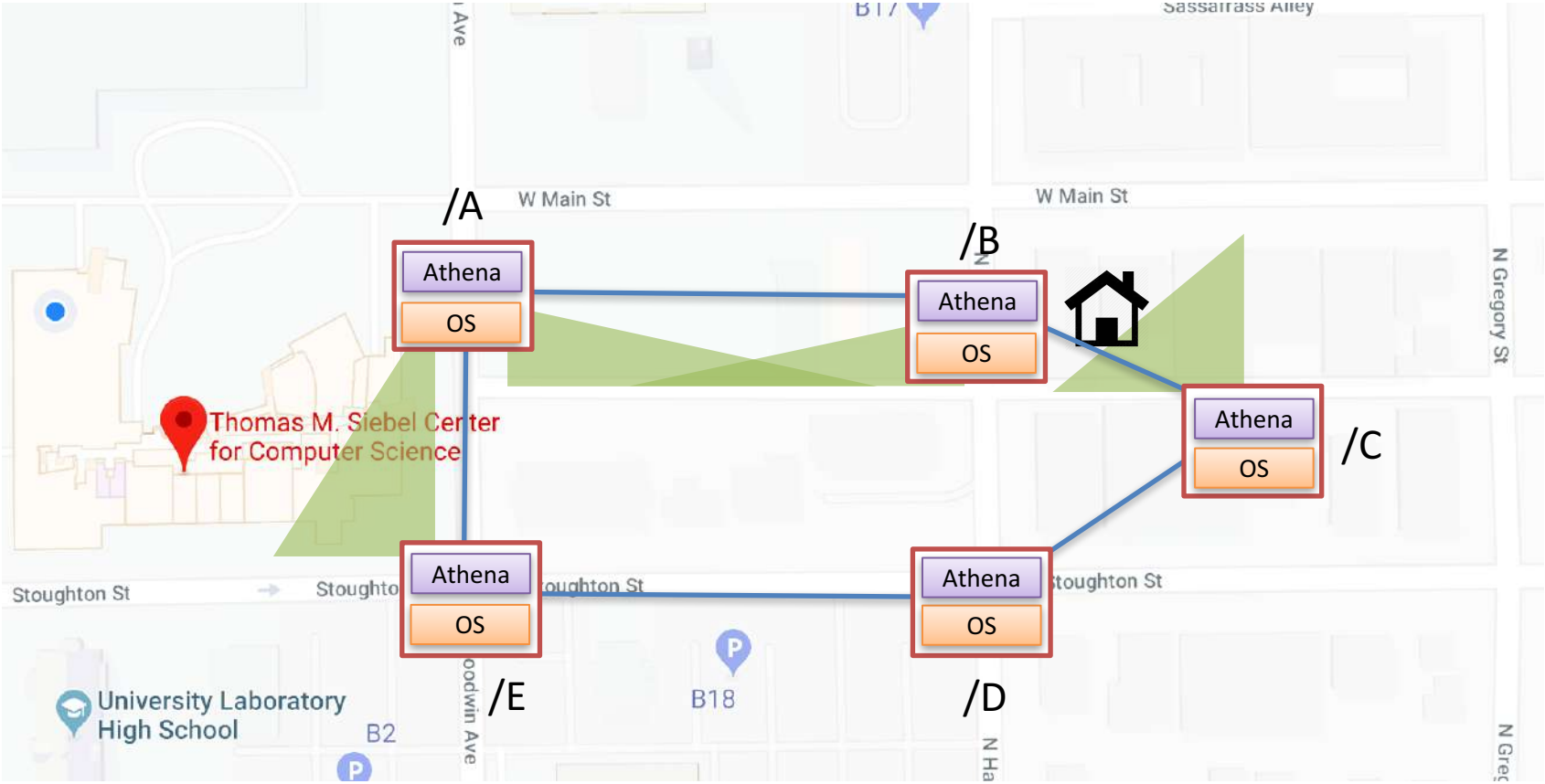


Athena Framework

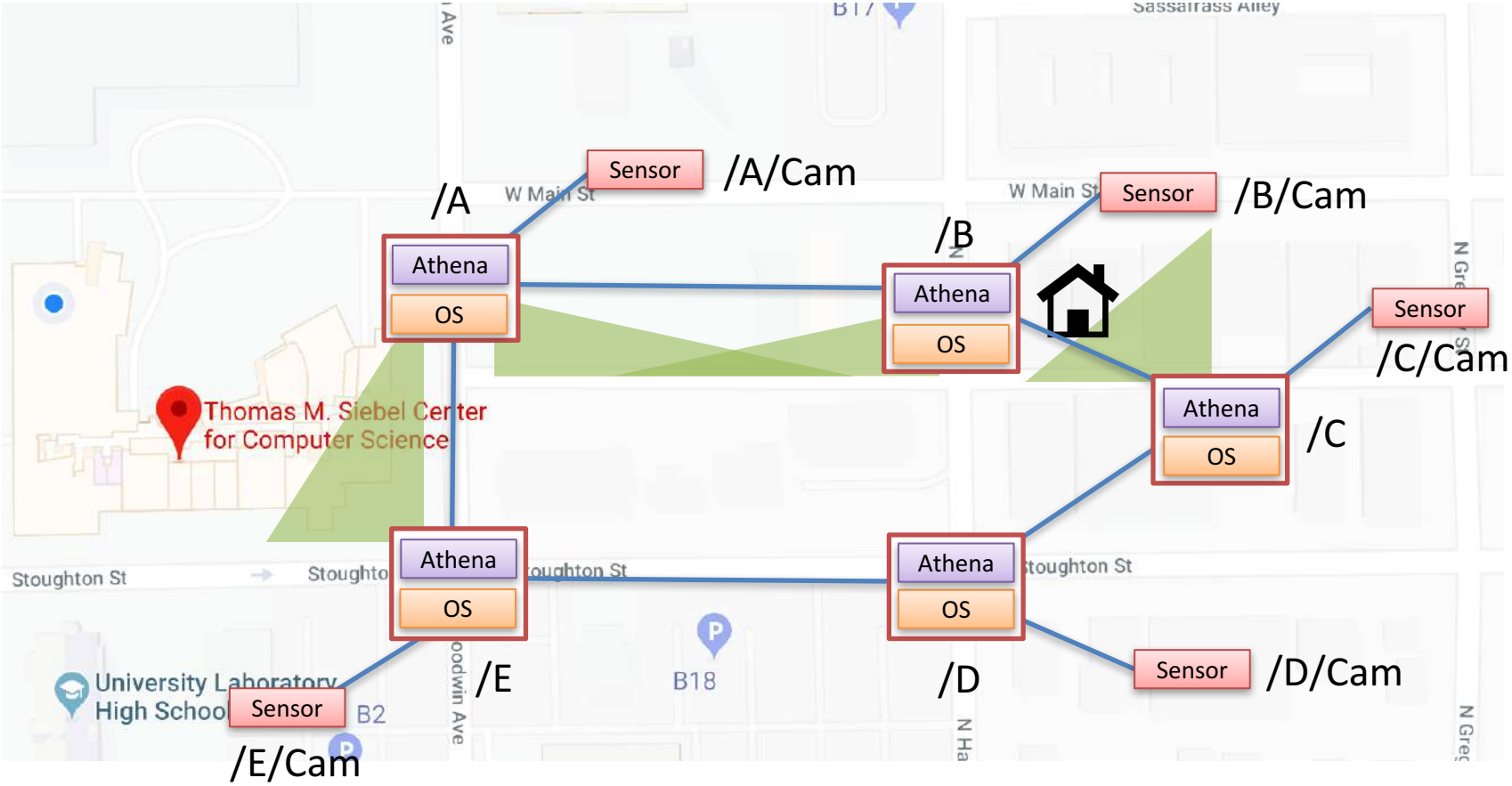
Athena Framework

- **Goal: Implement decision-driven execution**
  - Distributed system
  - Modular components
  - APIs for developers
  - Name-based data retrieval

Athena Framework



Athena Framework



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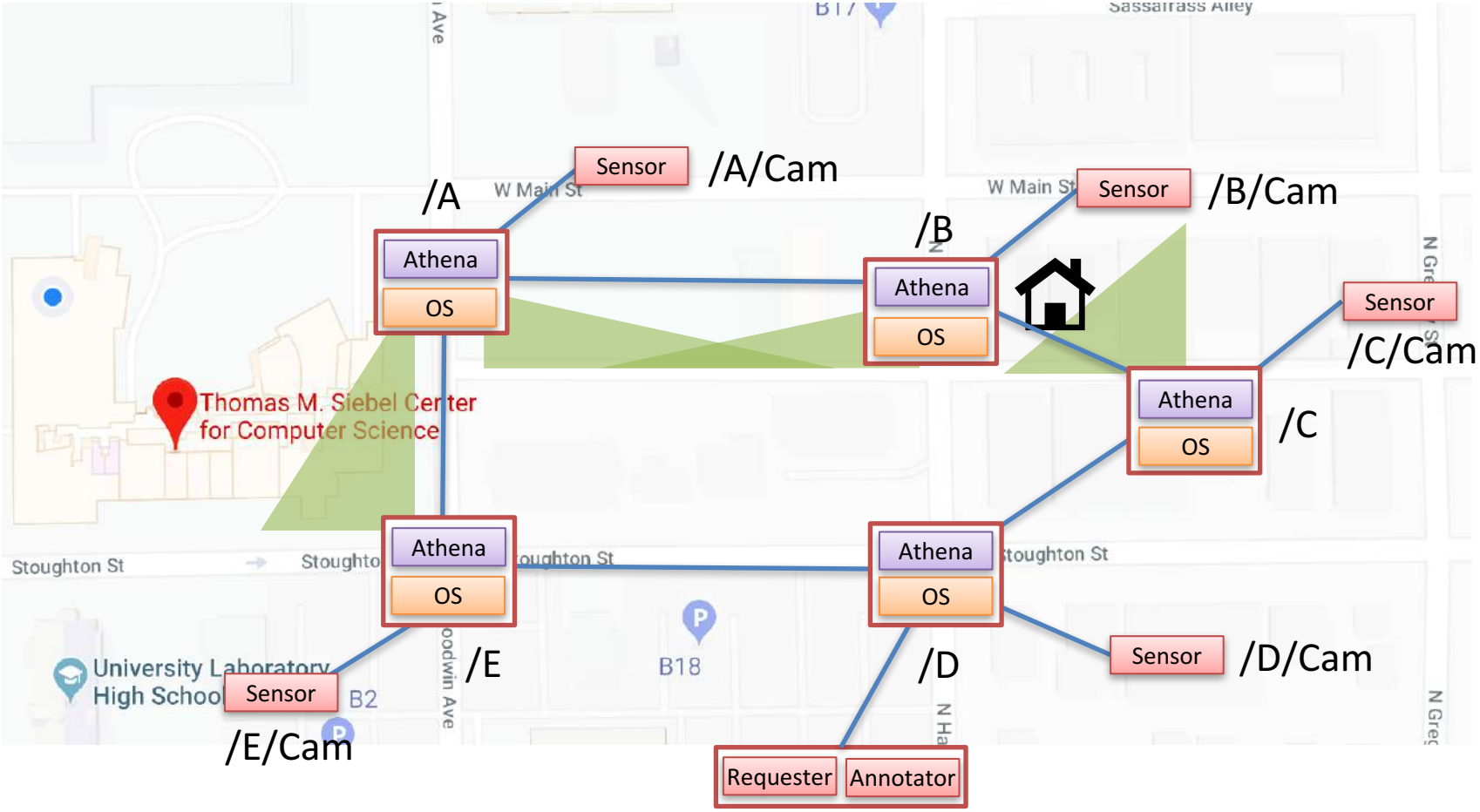

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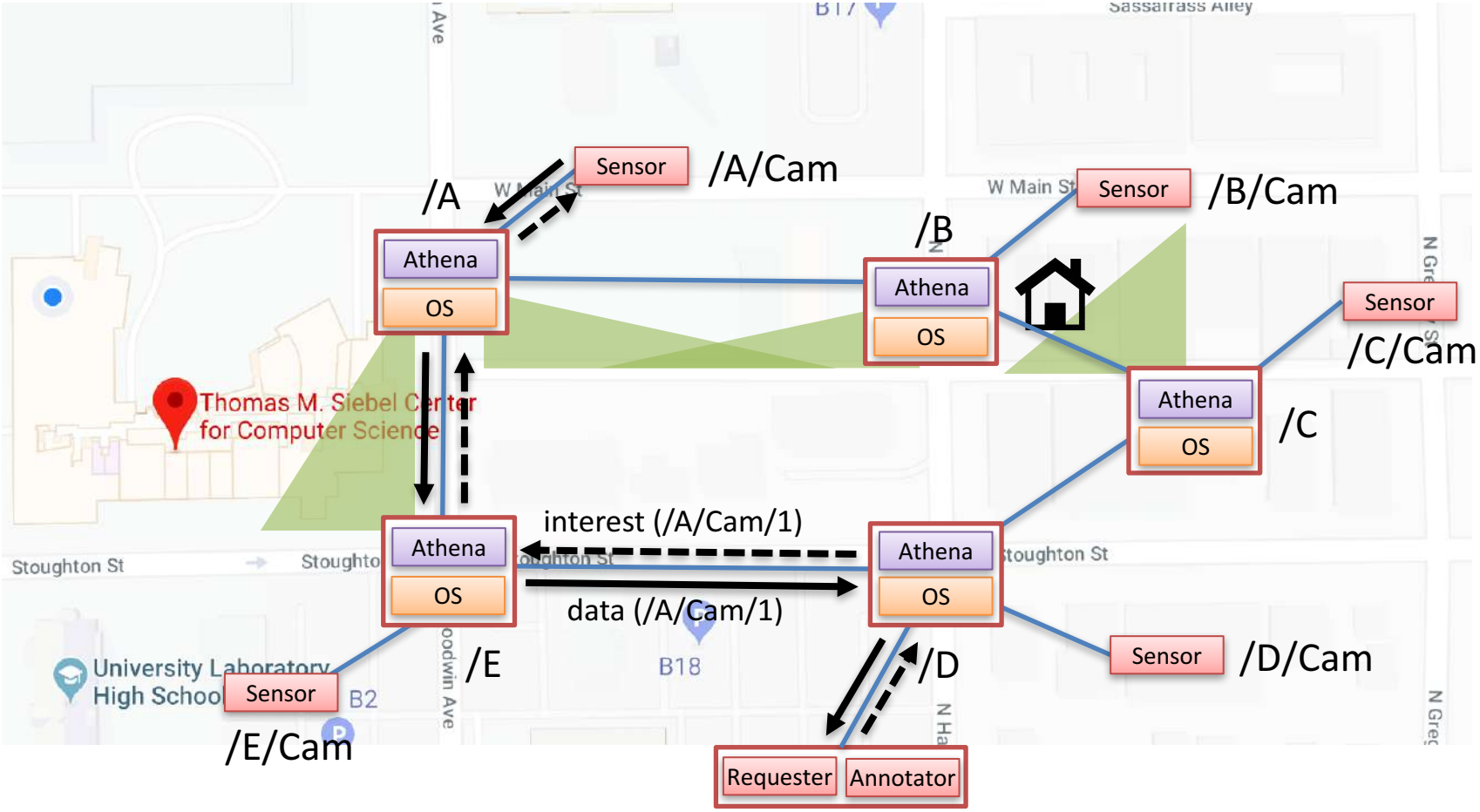

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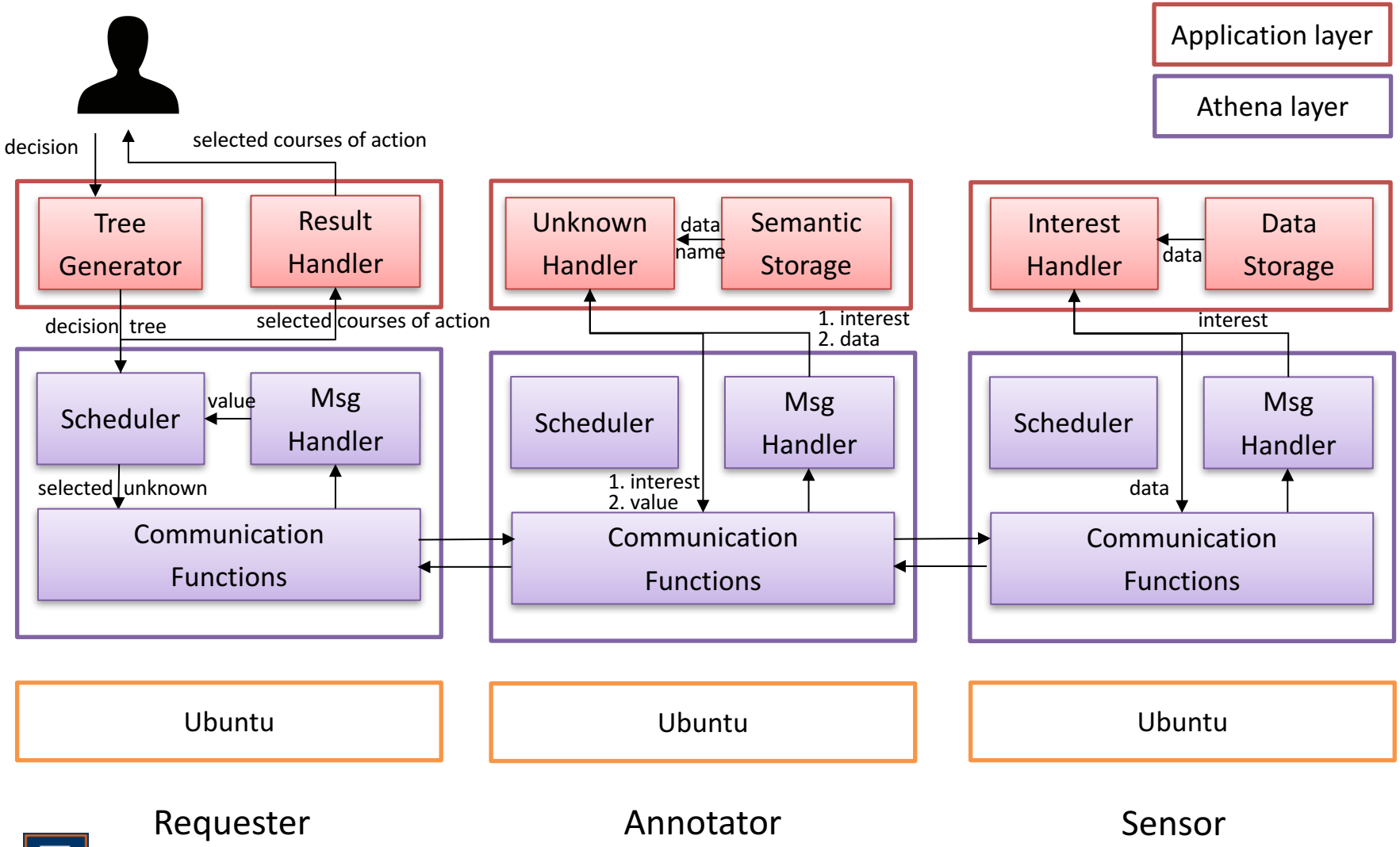


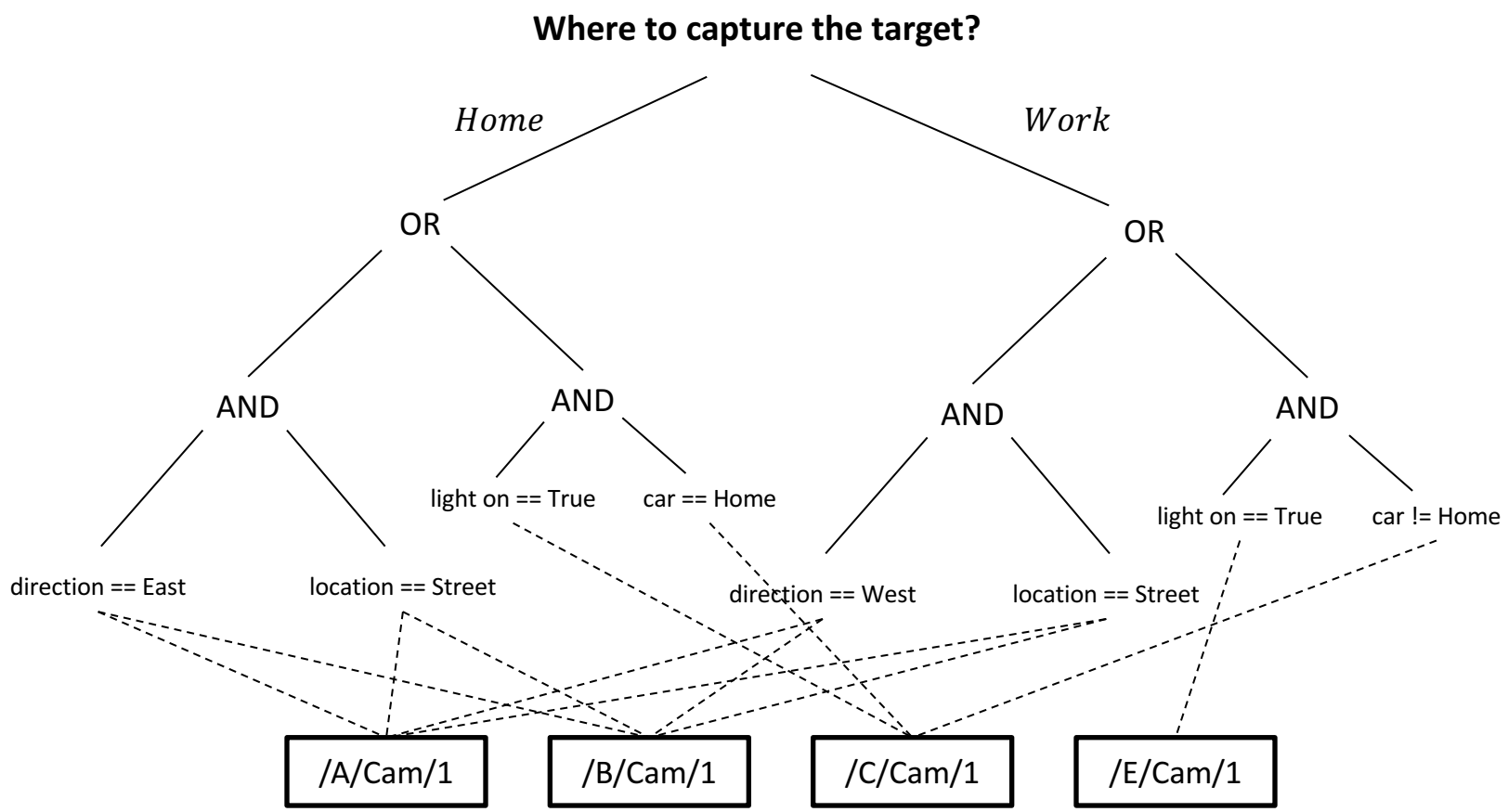
Prediction-based Optimization

Predication-based Optimization

- **Goal: Optimize data retrieval scheduling by anticipating the future state of physical environment**
  - Exploit short-circuit evaluation
  - Application-specific prediction model

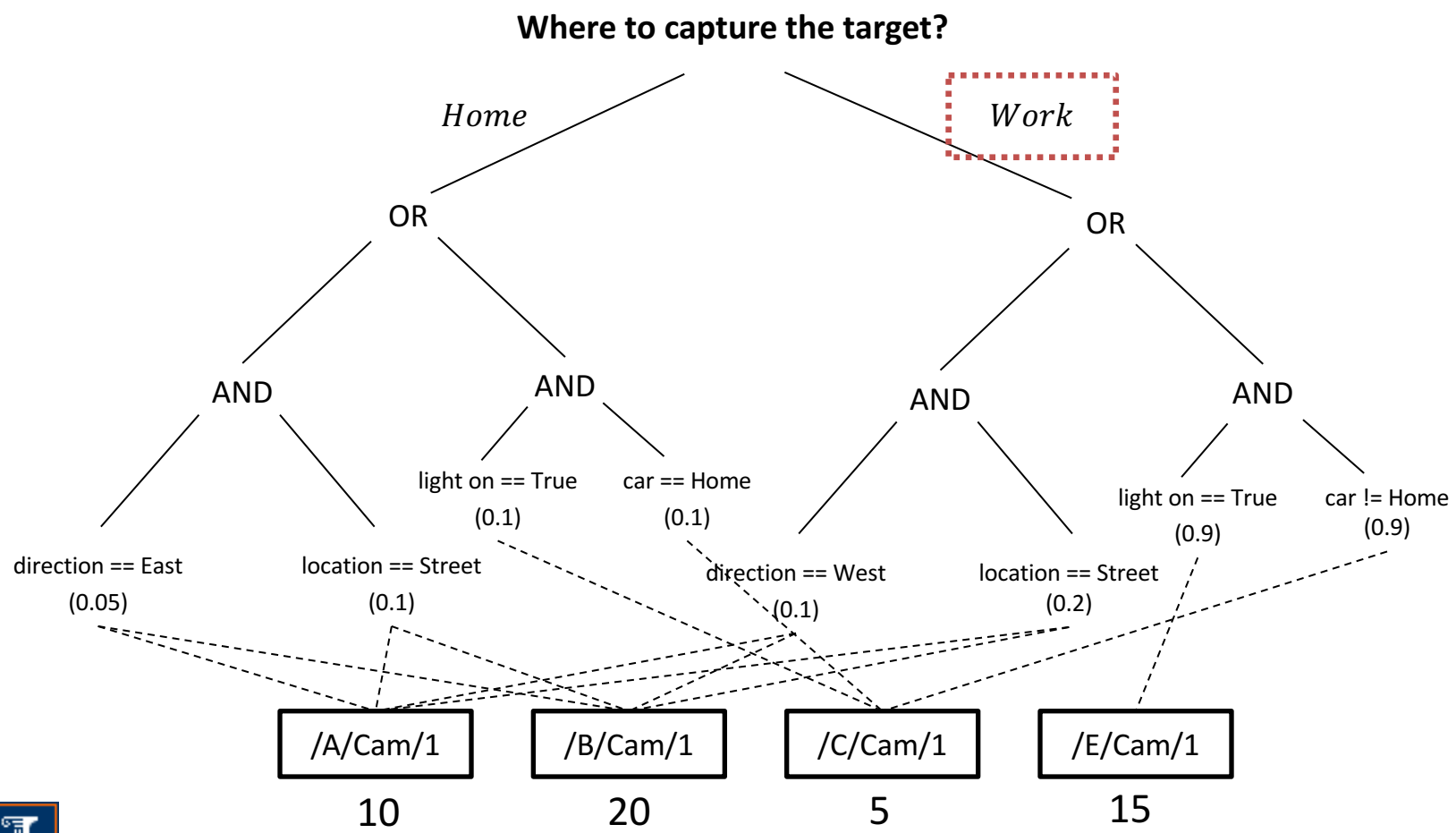
Prediction-based Optimization

- Daytime vs. Nighttime



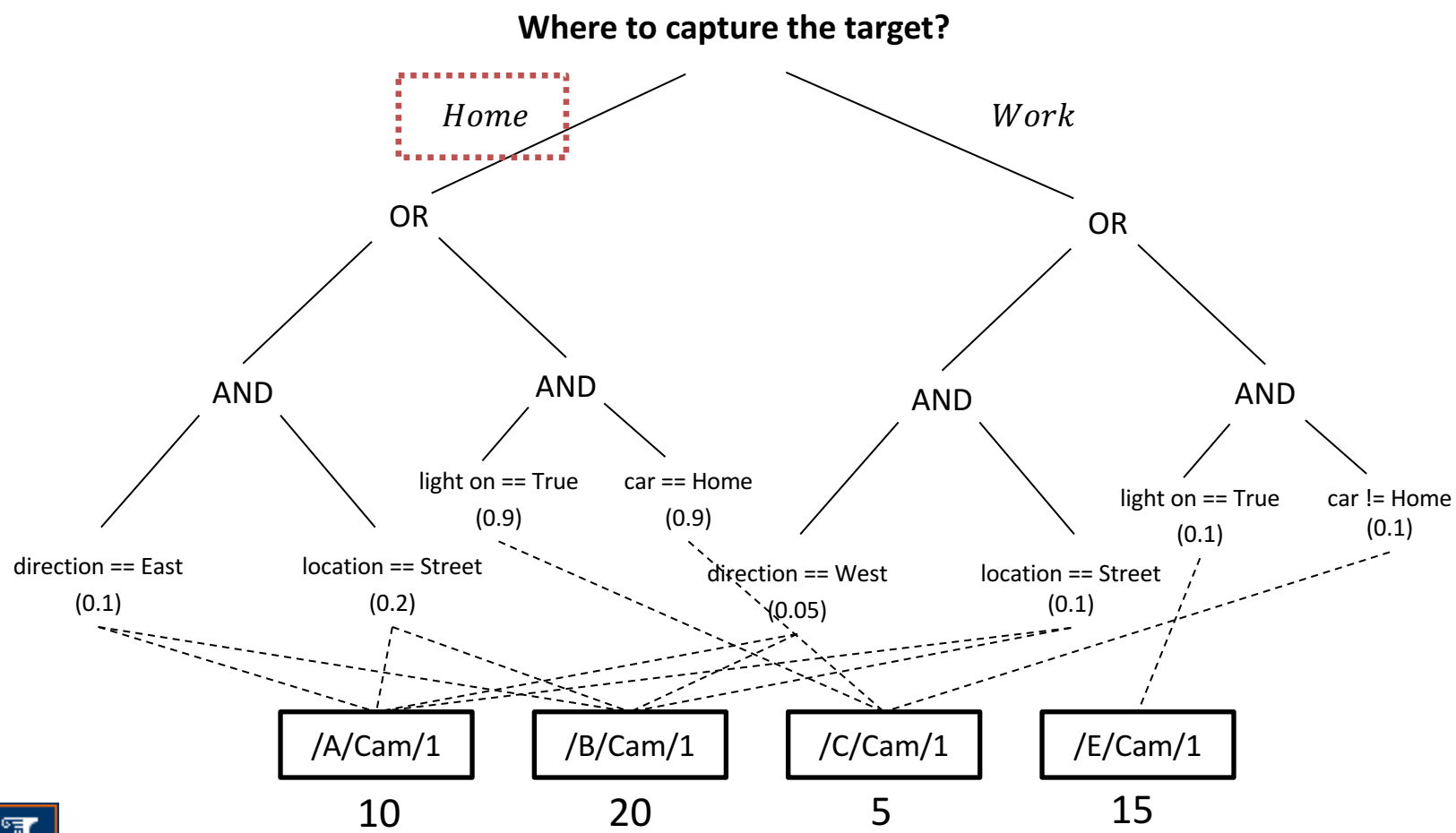
Prediction-based Optimization

- Probability and cost for **daytime**



Prediction-based Optimization

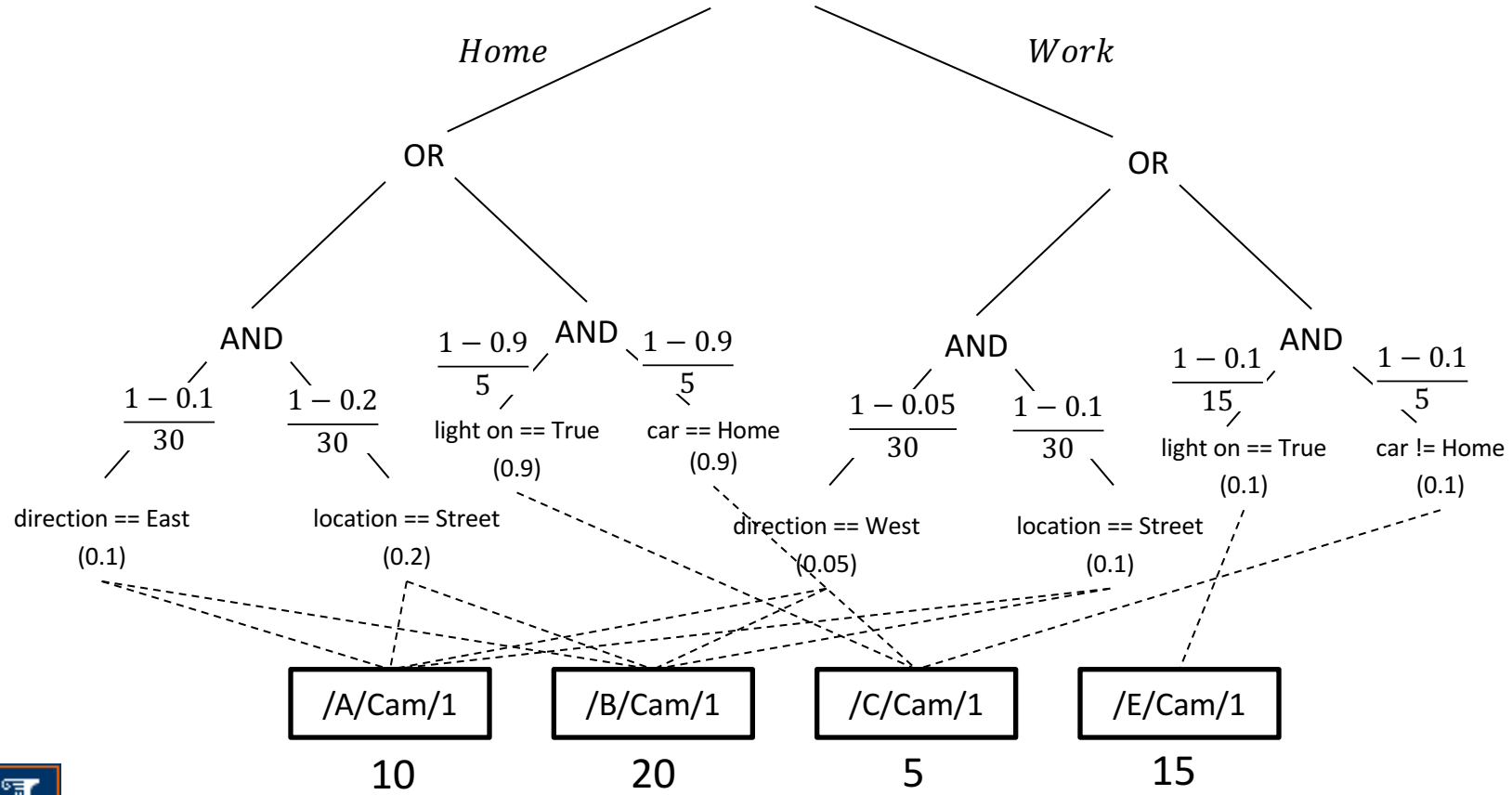
- Probability and cost for **nighttime**



Short-circuit Evaluation

- FALSE probability per unit cost for leaf nodes (C_{ijk}): $\frac{1-p_{ijk}}{c_{ijk}}$

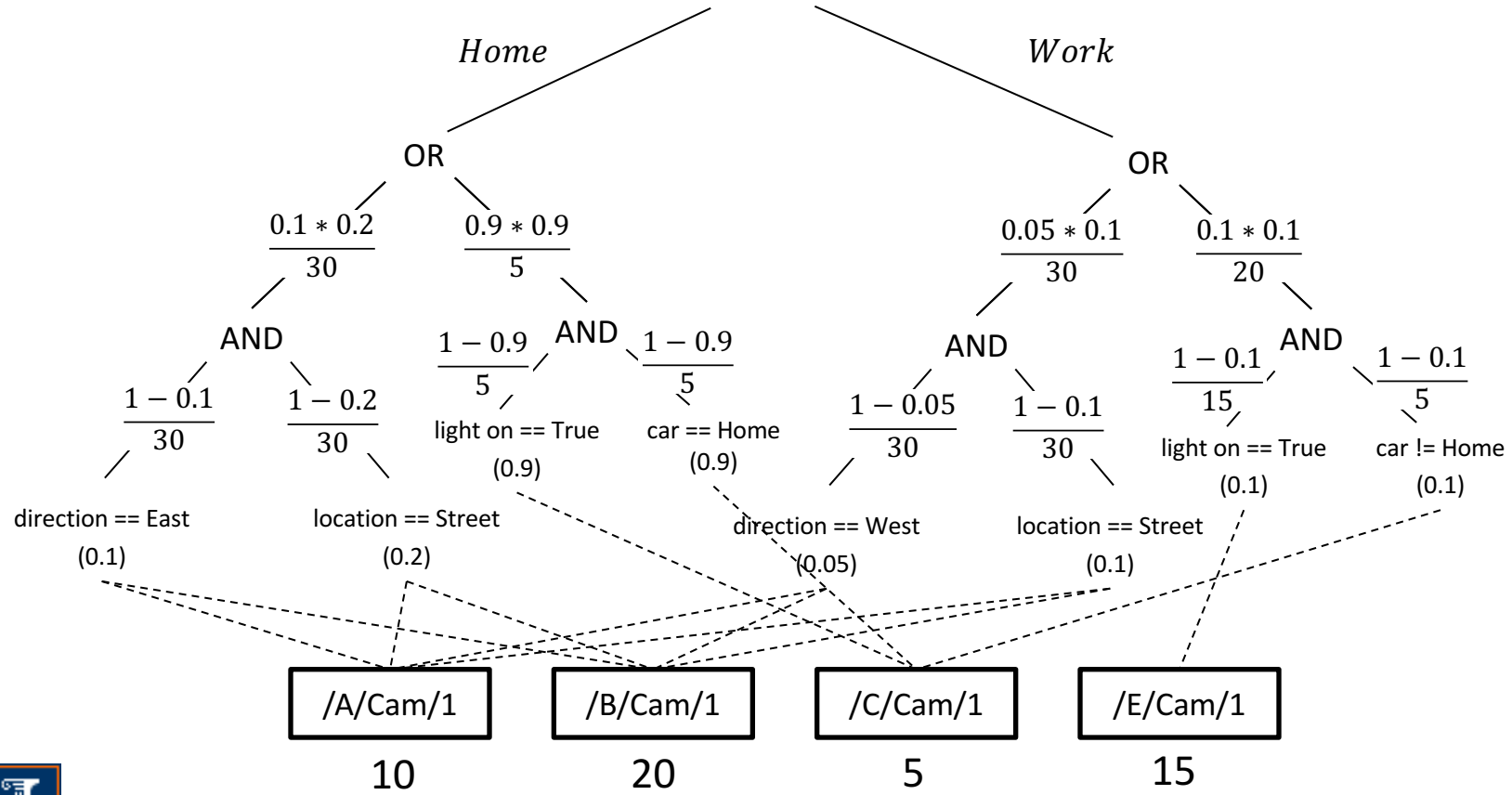
Where to capture the target?



Short-circuit Evaluation

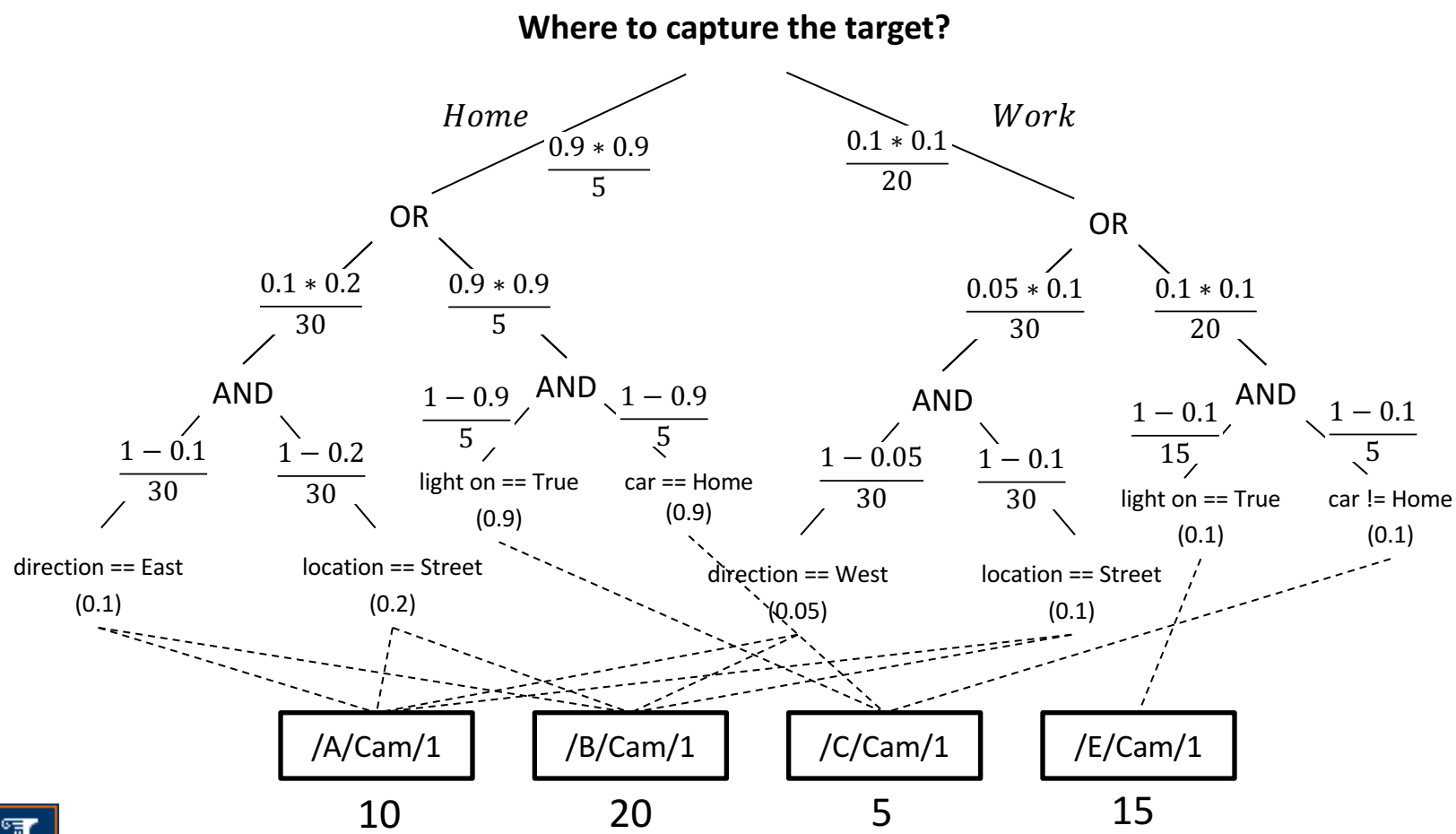
- TRUE probability per unit cost for AND nodes (C_{ij}): $\frac{\prod p_{ijk}}{\sum c_{ijk}}$

Where to capture the target?



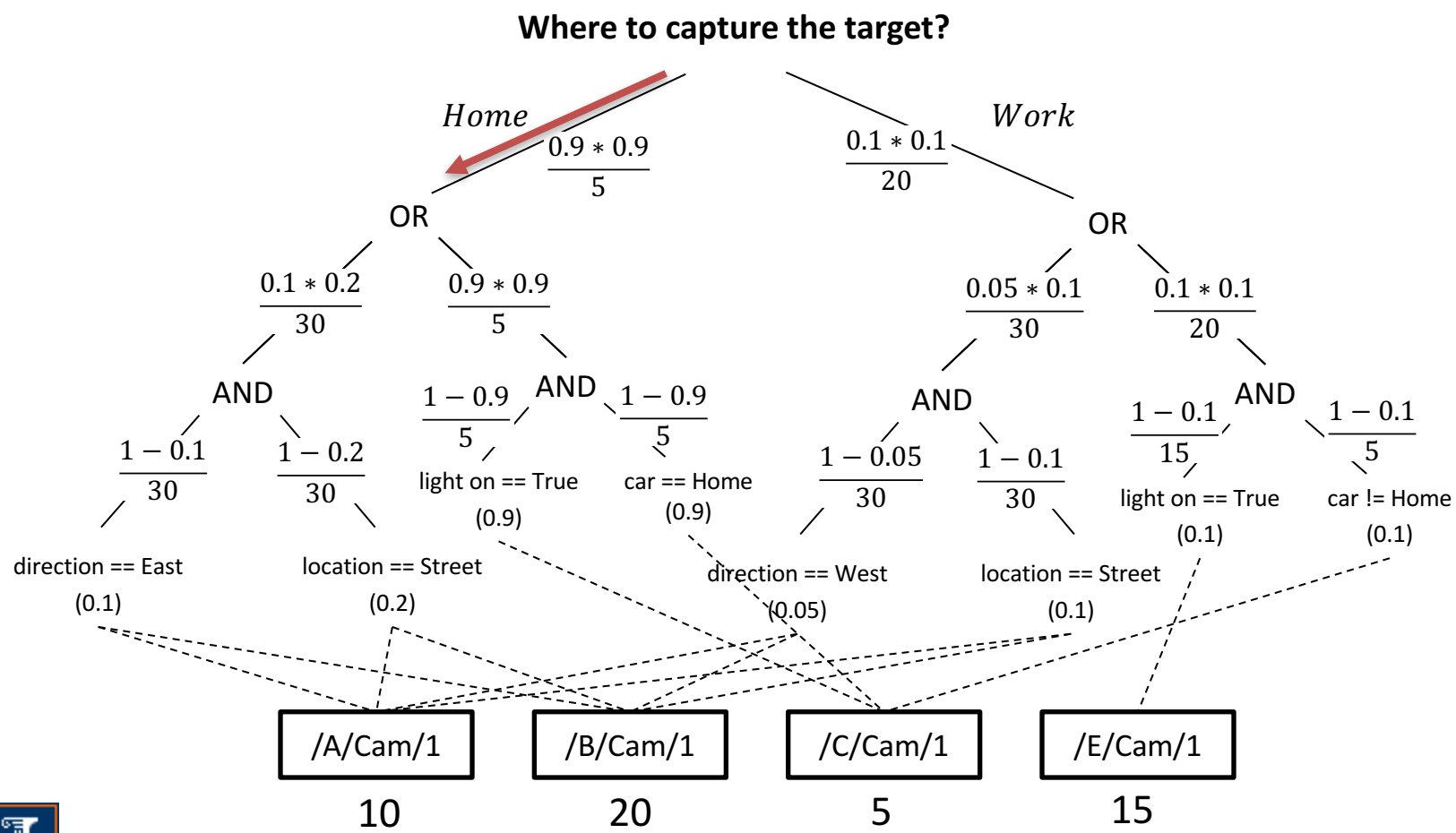
Short-circuit Evaluation

- Max TRUE probability per unit cost for OR nodes (C_i): $\max(C_{i1}, C_{i2}, \dots, C_{ij})$



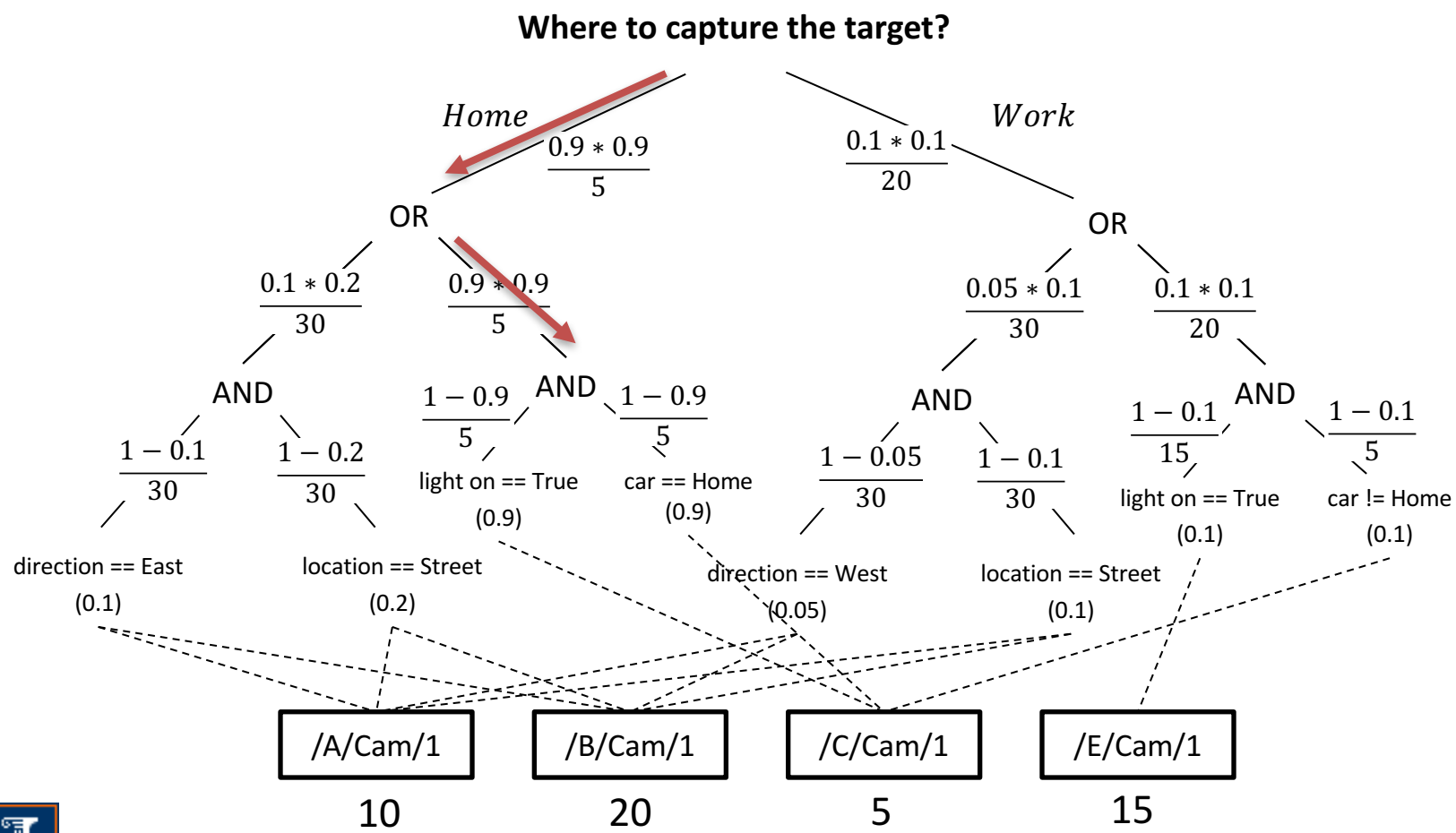
Short-circuit Evaluation

- Select the courses of action that is more likely true with minimal cost



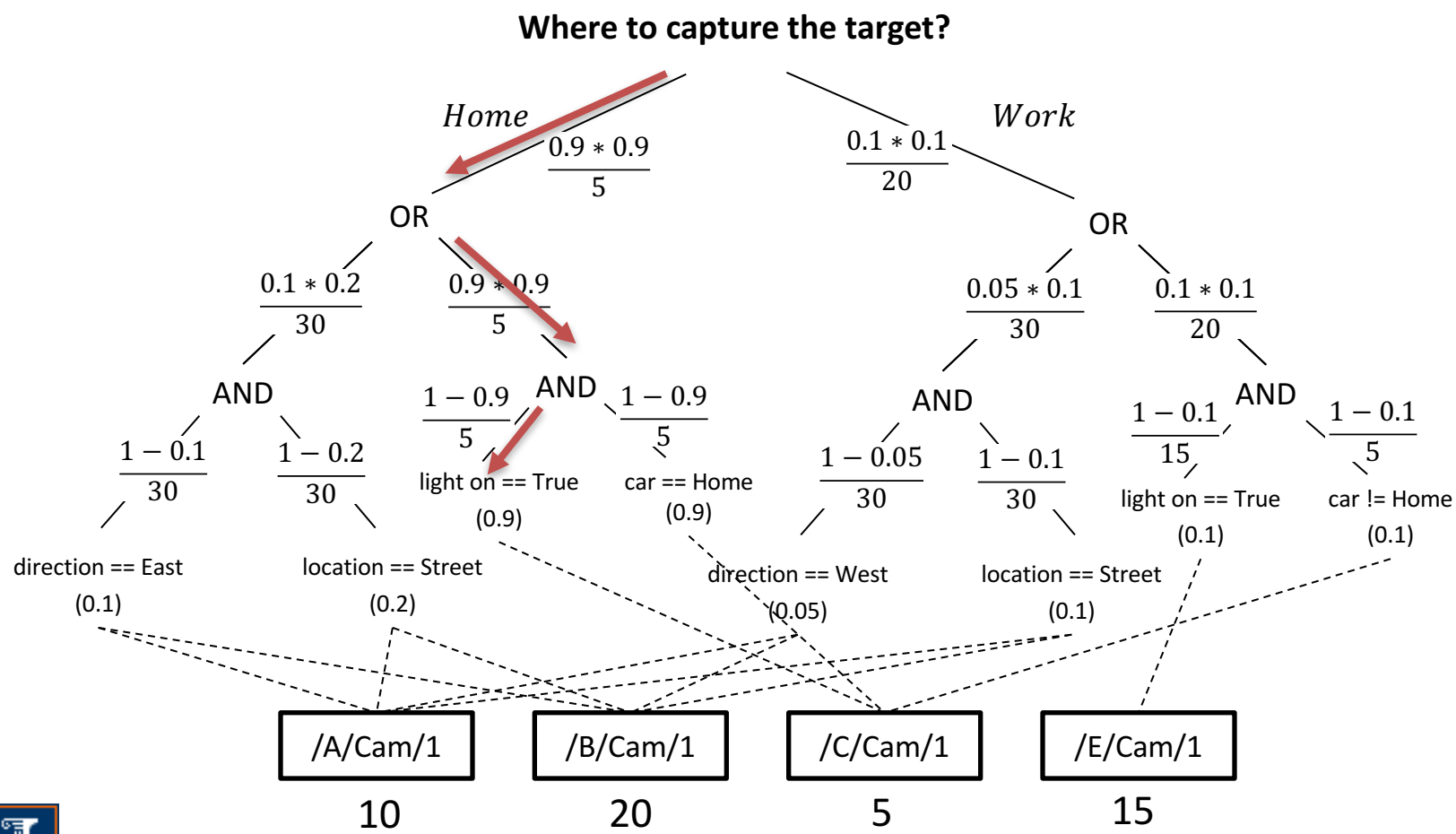
Short-circuit Evaluation

- Select the AND branch that is more likely true with minimal cost



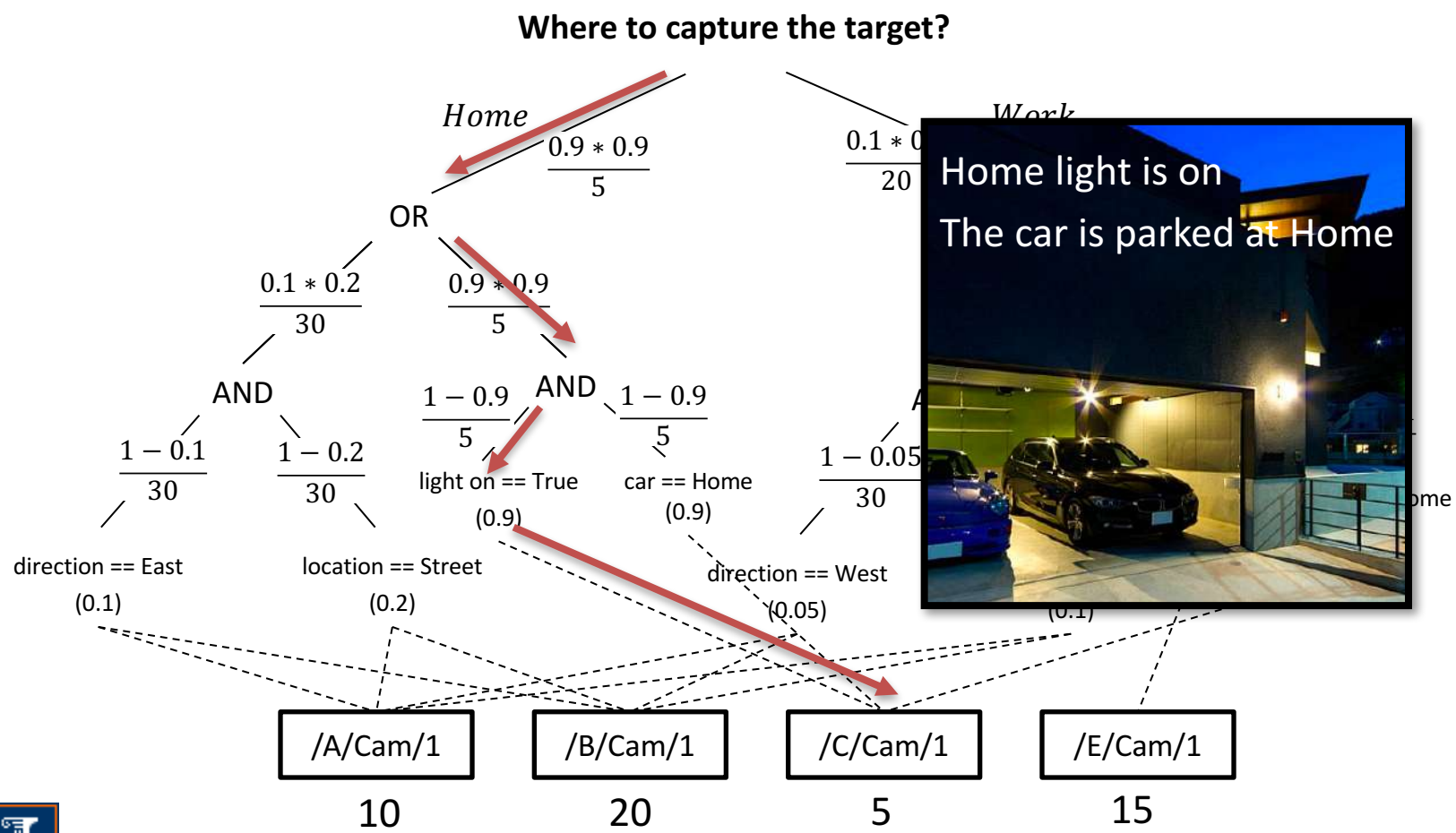
Short-circuit Evaluation

- Select the predicate that is more likely false with minimal cost



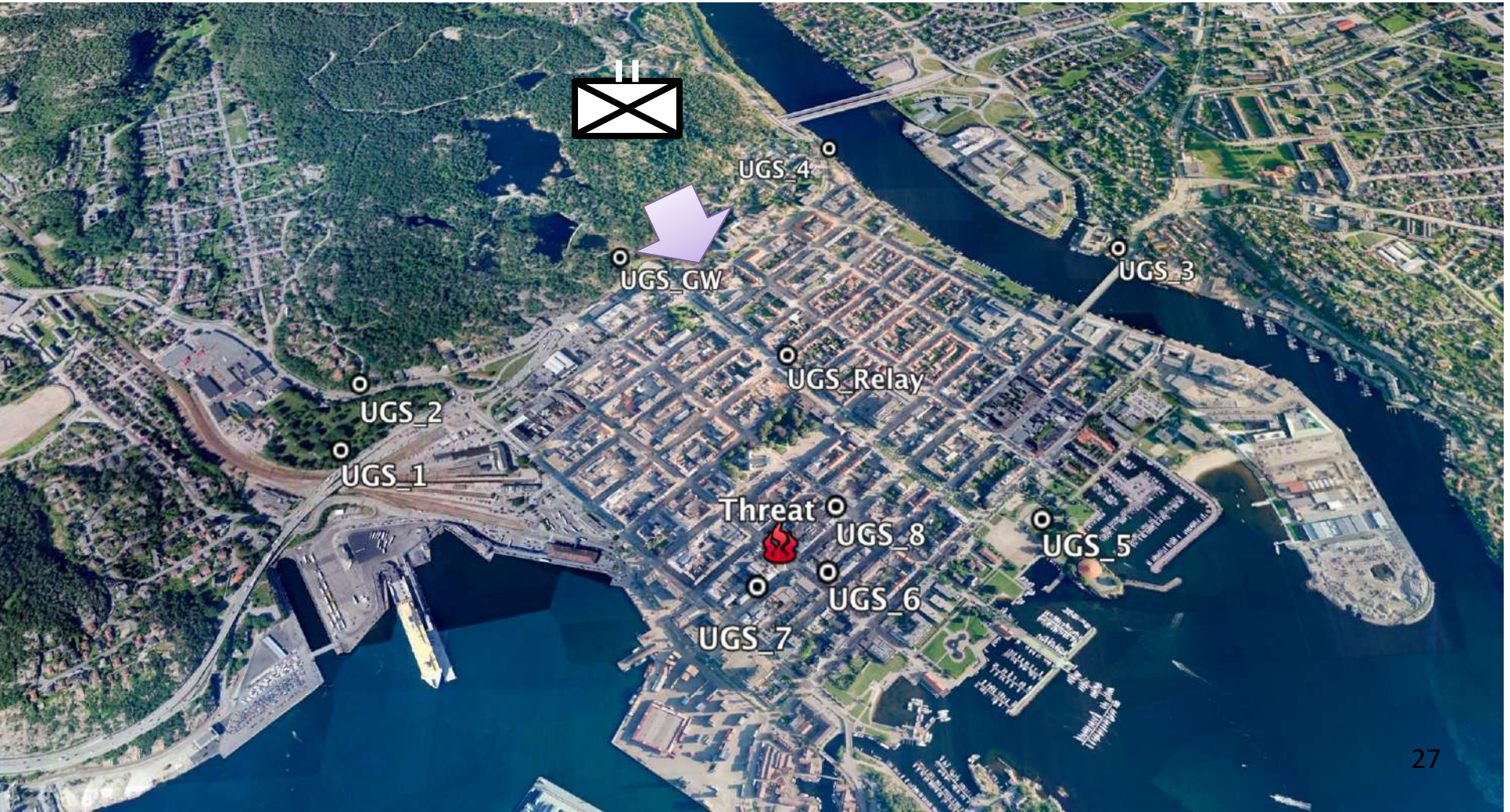
Short-circuit Evaluation

- Retrieve data that offers evidence for the selected unknown



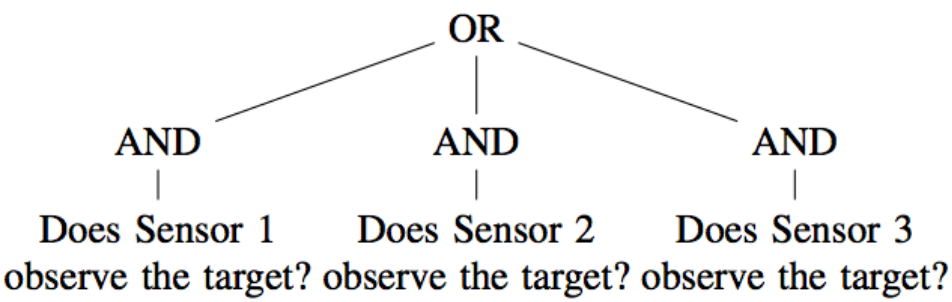
Evaluation Setup

- **Anglova scenario**



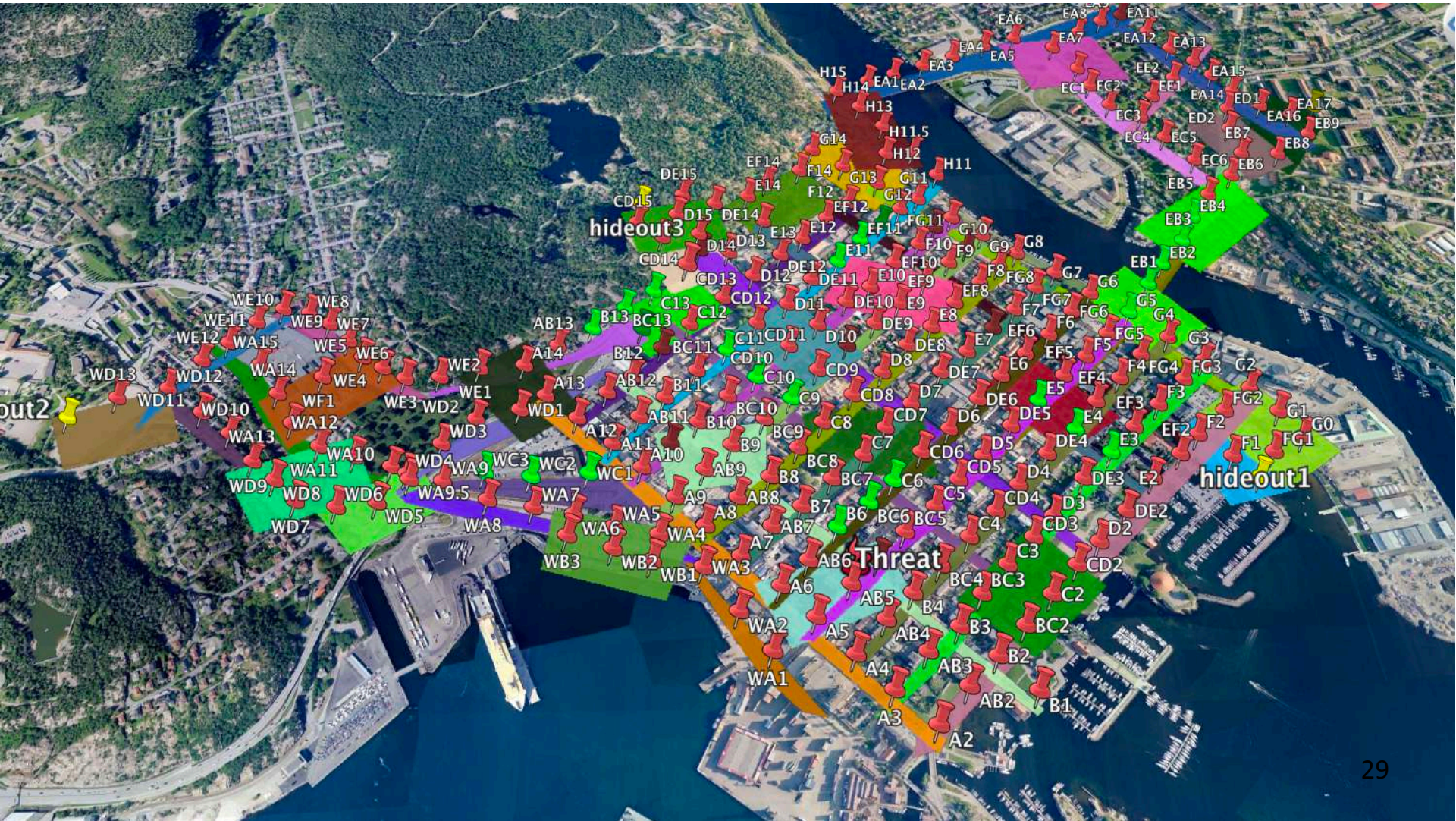
Evaluation Setup

- **Decision model**
  - Decision: Where to capture the target?
  - Courses of actions: point of interests
    - » Ex) EF4, F3, F4, F5, FG4, …
  - Validity logic
    - » Capture the target at the POI if the target is at the POI



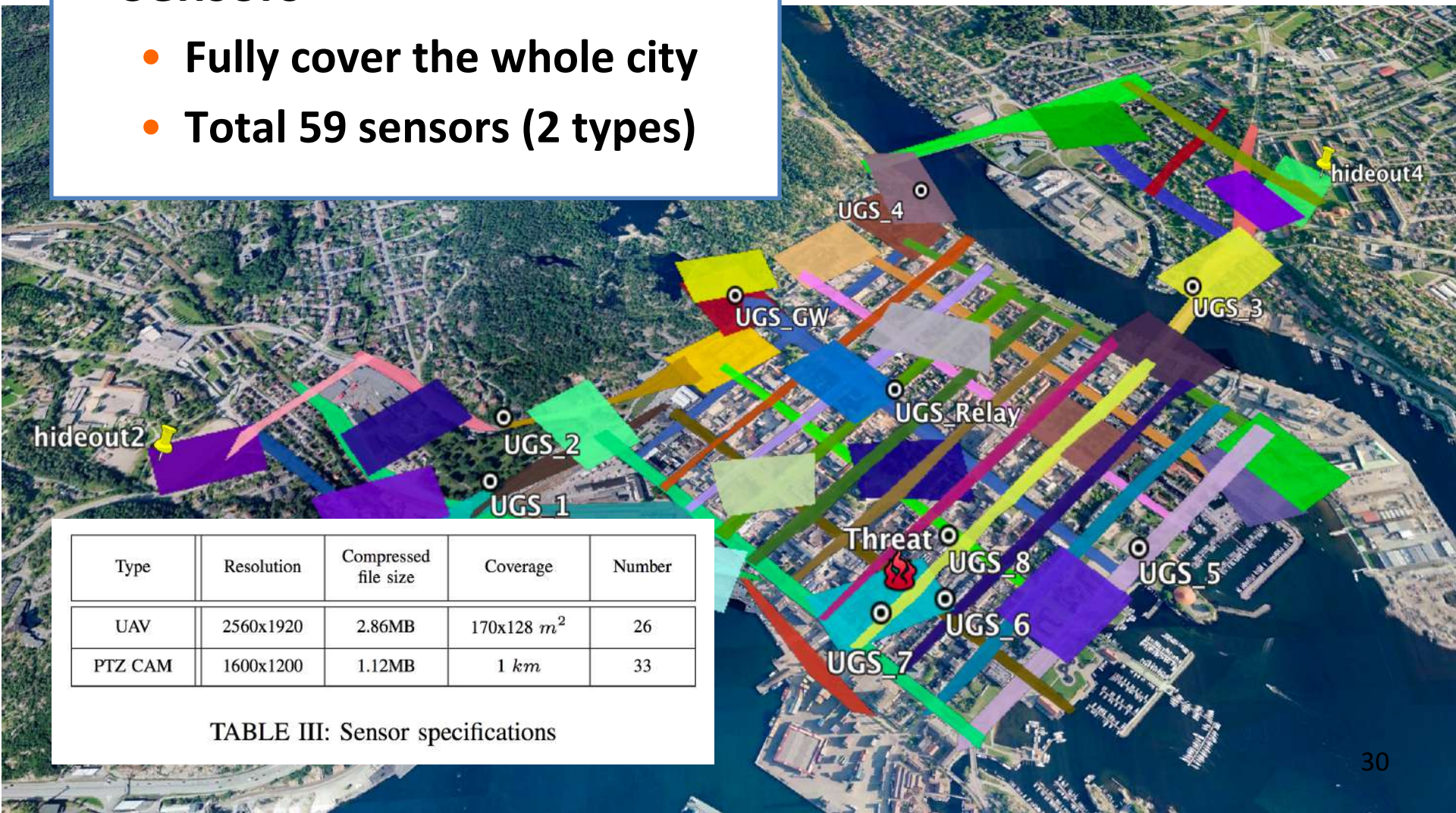
Evaluation Setup

- **POIs**



Evaluation Setup

- **Sensors**
  - **Fully cover the whole city**
  - **Total 59 sensors (2 types)**



Type	Resolution	Compressed file size	Coverage	Number
UAV	2560x1920	2.86MB	170x128 m^2	26
PTZ CAM	1600x1200	1.12MB	1 km	33

TABLE III: Sensor specifications

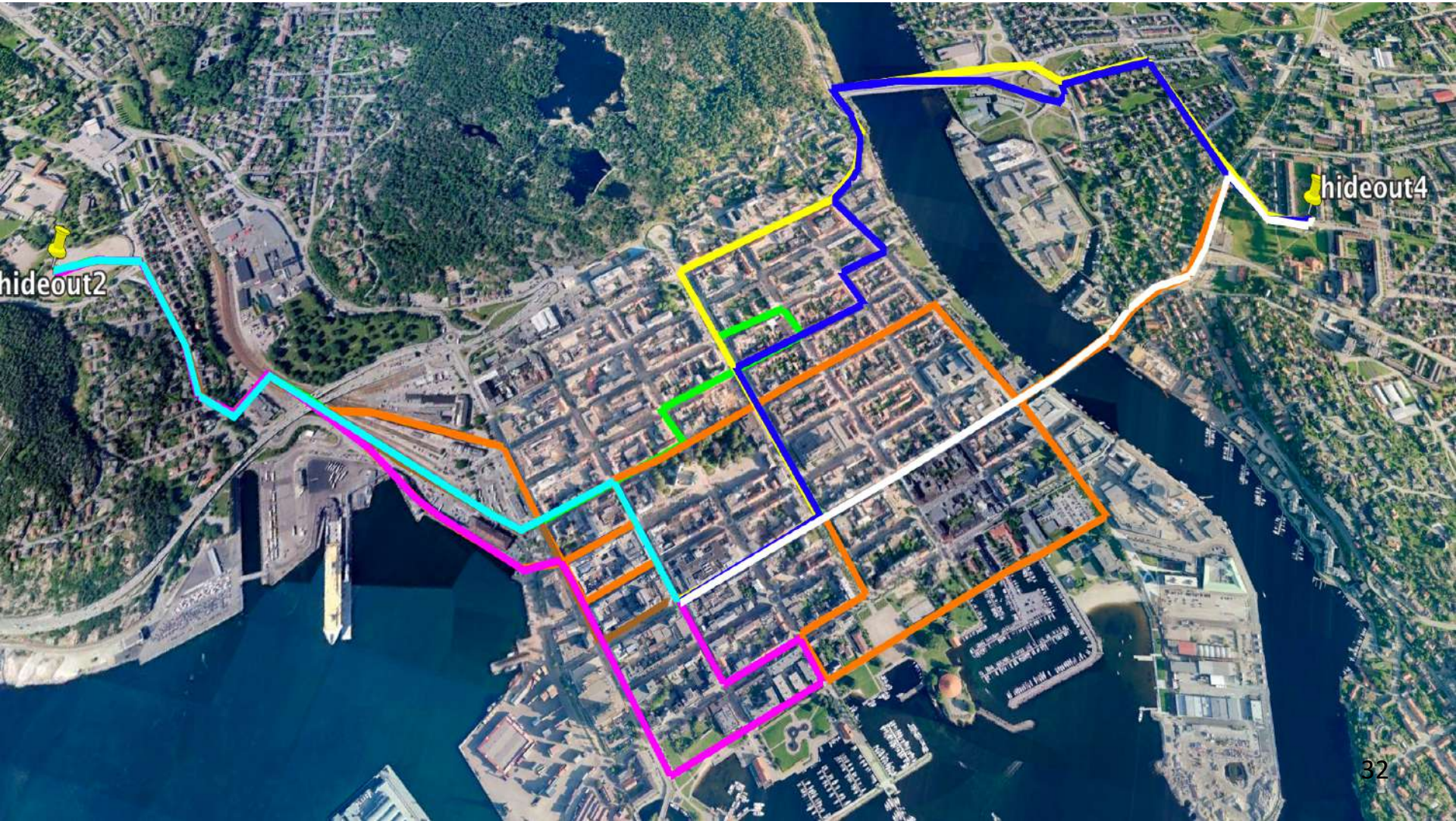
Evaluation Setup

- **Targets**
  - Vehicle types: car / auto-bike
  - Hideout points: west / east
  - Mobility pattern: simple / deceptive

- **Anticipation model**
  - Compute statistics for simulated targets
    - » Given the current position, what is the probability that the target moves to the next position?
    - » ex) f('A7', 'C7', car, west, simple) = 0.3

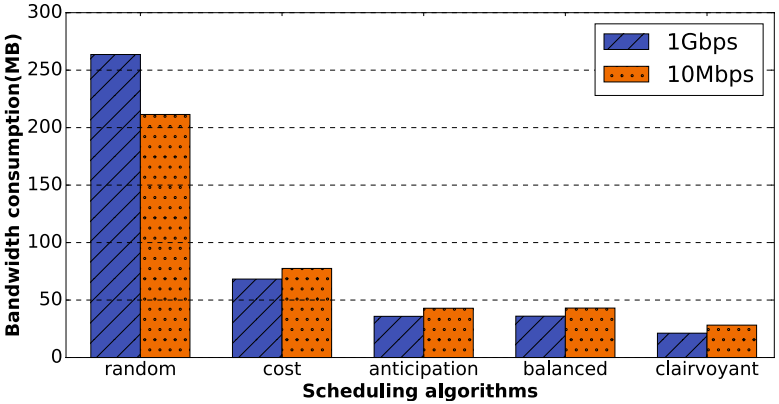
Evaluation Setup

- **Targets**

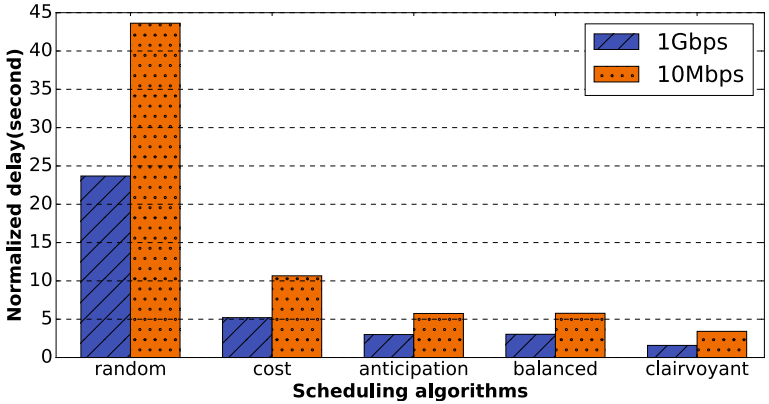


Evaluation Results

- **Scheduling algorithms**



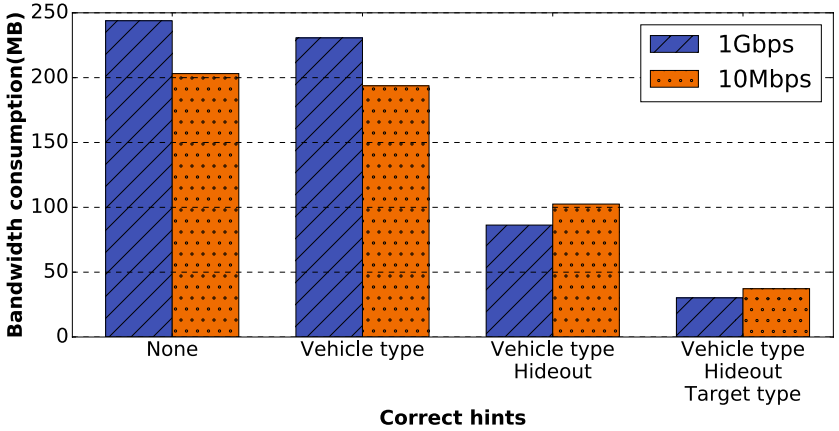
Bandwidth consumption of a requester node



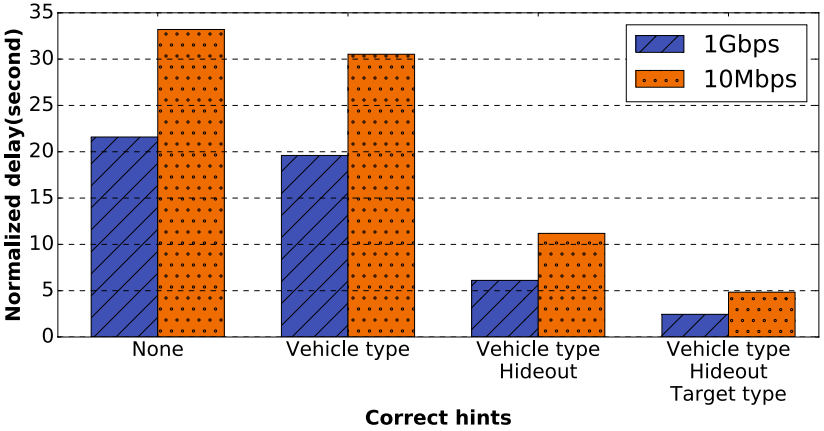
Query resolution delay

Evaluation Results

- **Anticipation accuracy**



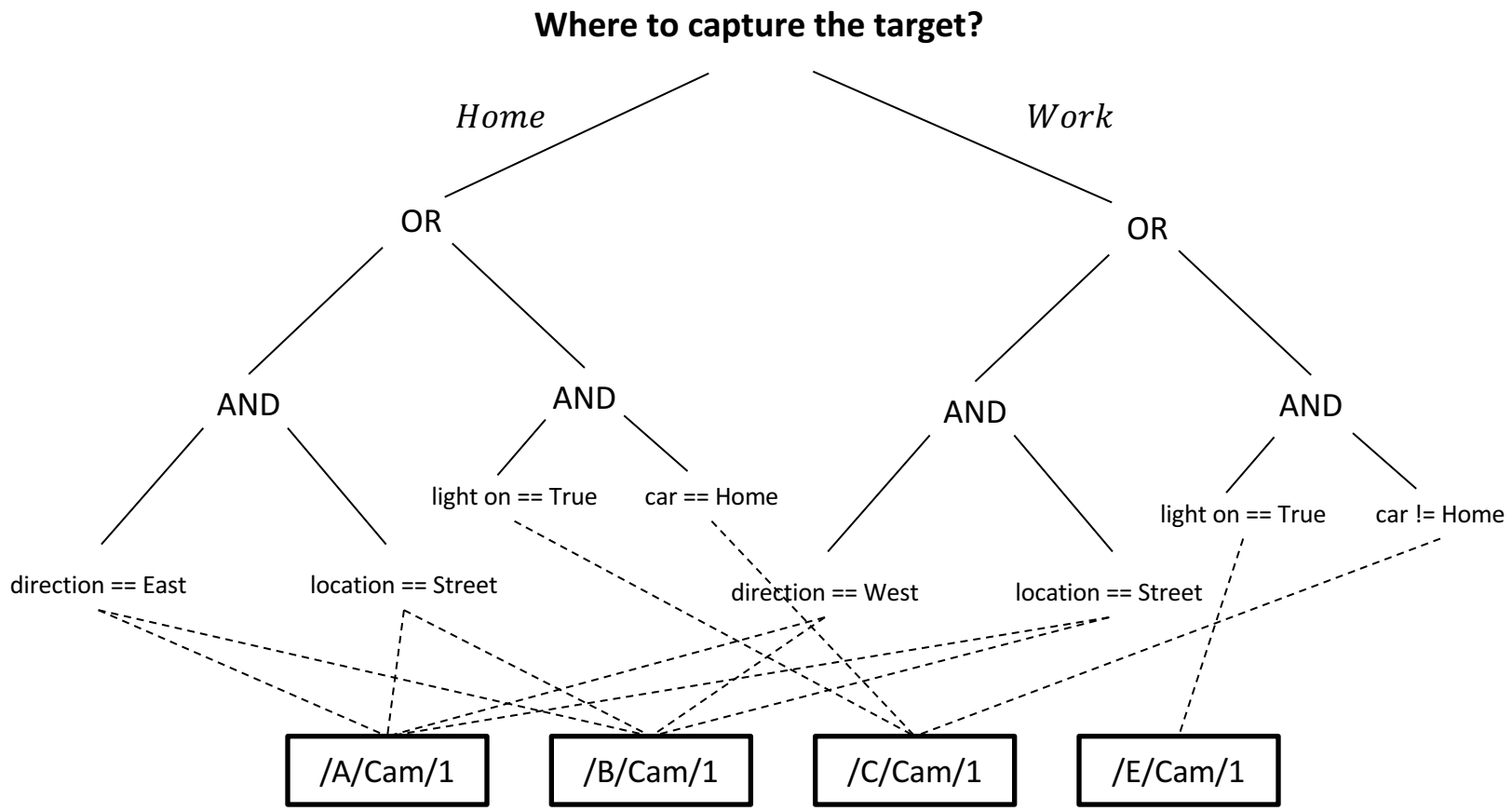
Bandwidth consumption of a requester node



Query resolution delay

Challenge

- **Data vs. Data set**



Challenge

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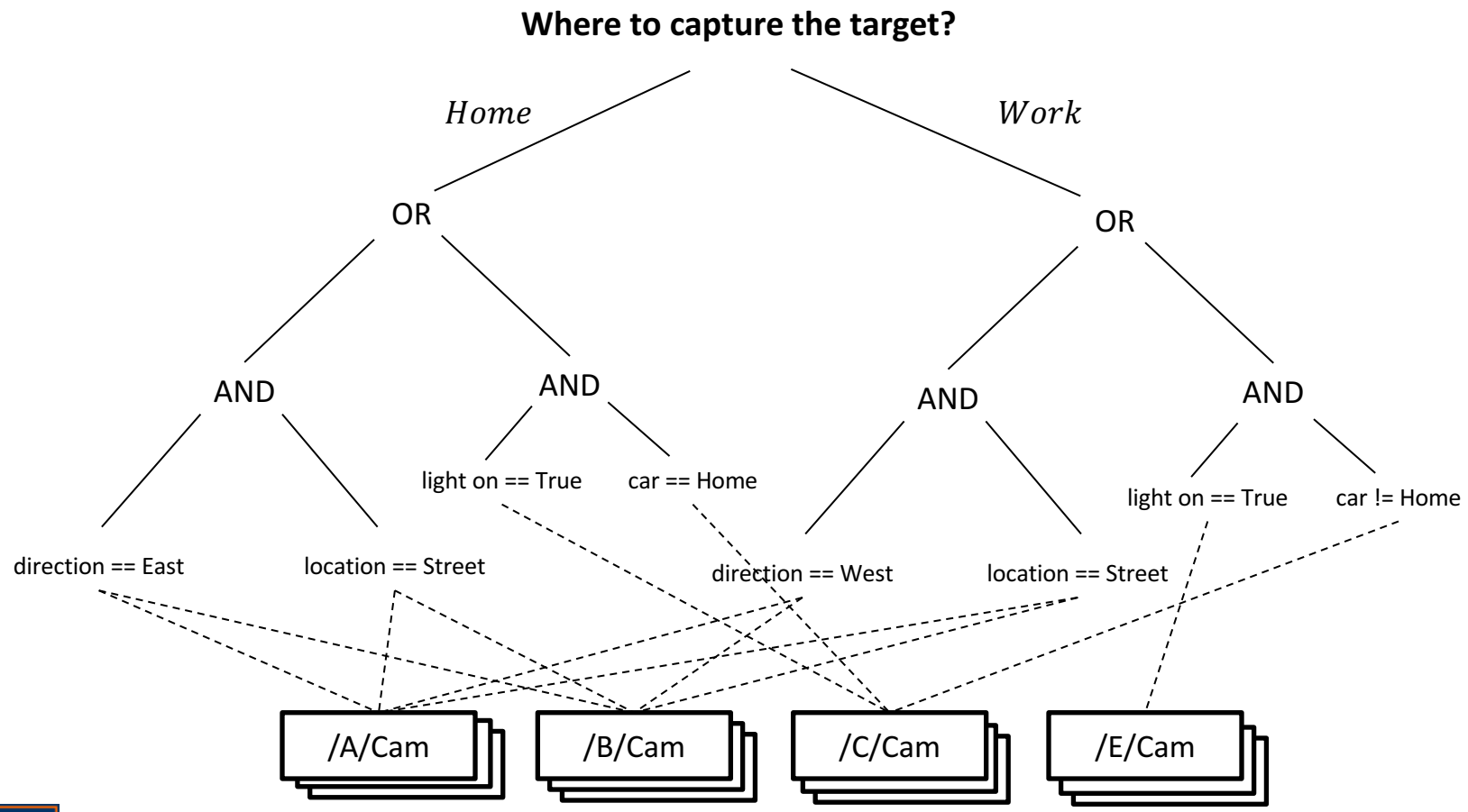


Redundancy-based Optimization

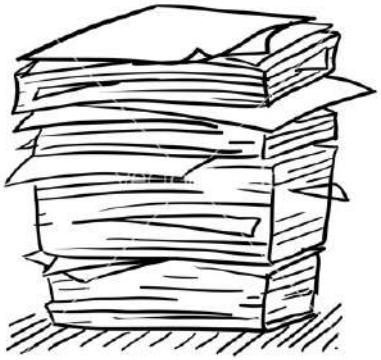
Redundancy-based Optimization

- **Goal: Design and implement a content producer and consumer that jointly offer a content sampling service**
  - InfoMax: **Info**rmation **Max**imizing data retrieval (maximizes "information" given the number of objects retrieved)
  - A new protocol for retrieving **representative data samples** on top of the NDN stack

Data Sampling Service
- Extractive summarization –

Subscriber A

Request data tree rooted in /X



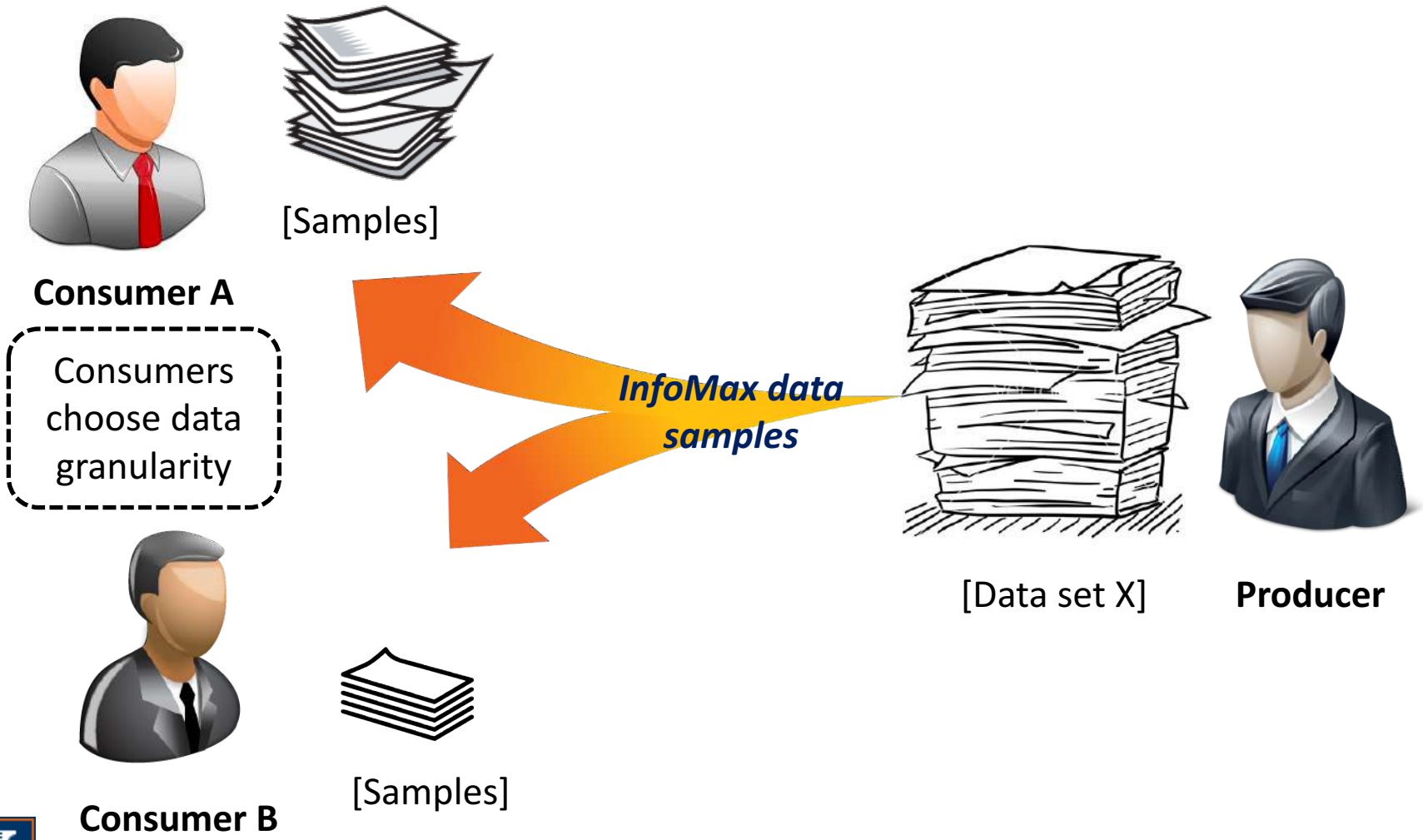
[Data set X]

Publisher

Request data tree rooted in /X

Subscriber B

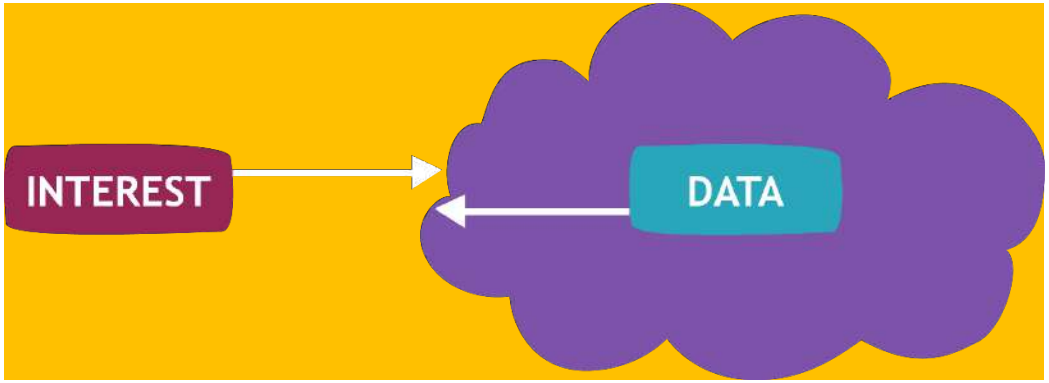
Data Sampling Service
- Extractive summarization –



Named Data Network (NDN)

Abstracting away the notion of "host"

Consumer



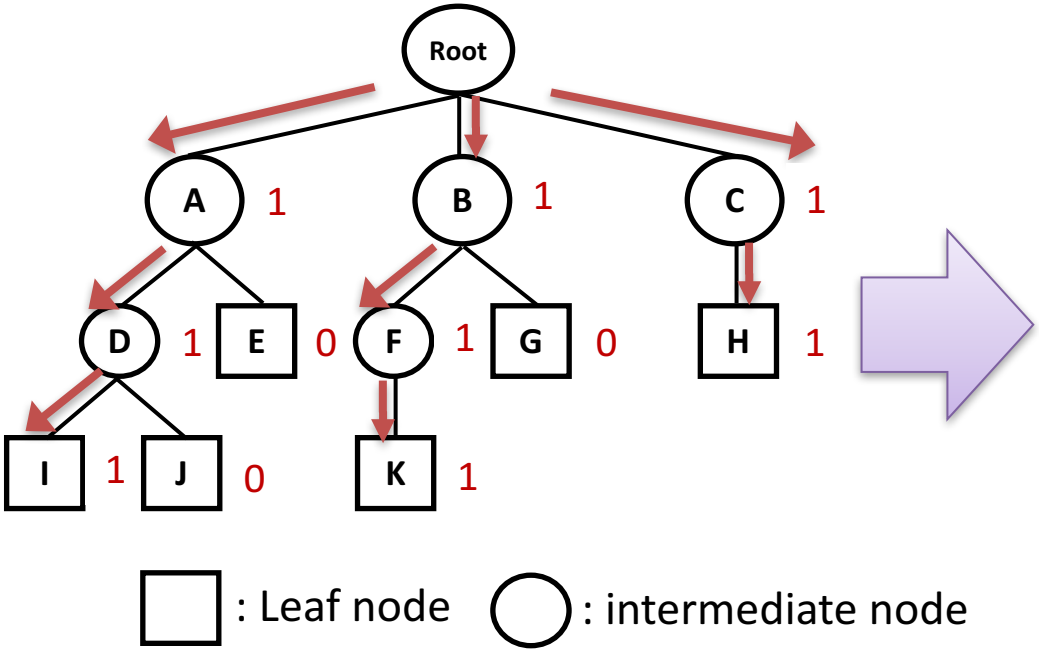
Producer

Unix-like contents name:
/youtube/video/friends/season1/ep1

InfoMax Order

Given a (Root) tree prefix:

- **Items that share longer prefix are more similar (redundant)**
- **Information maximizing order: minimum shared-prefix first**



Order	Name
1	/Root/A/D/I
2	/Root/B/F/K
3	/Root/C/H
4	/Root/A/E
5	/Root/B/G
6	/Root/A/D/J

Design Goals

- Consumers need not know entire application namespace a priori

➔ Producers create and maintain InfoMax order

- Enable multi-level summarization

➔ Bottom-up prioritization algorithm

- Maintain InfoMax delivery order in the presence of NDN caching

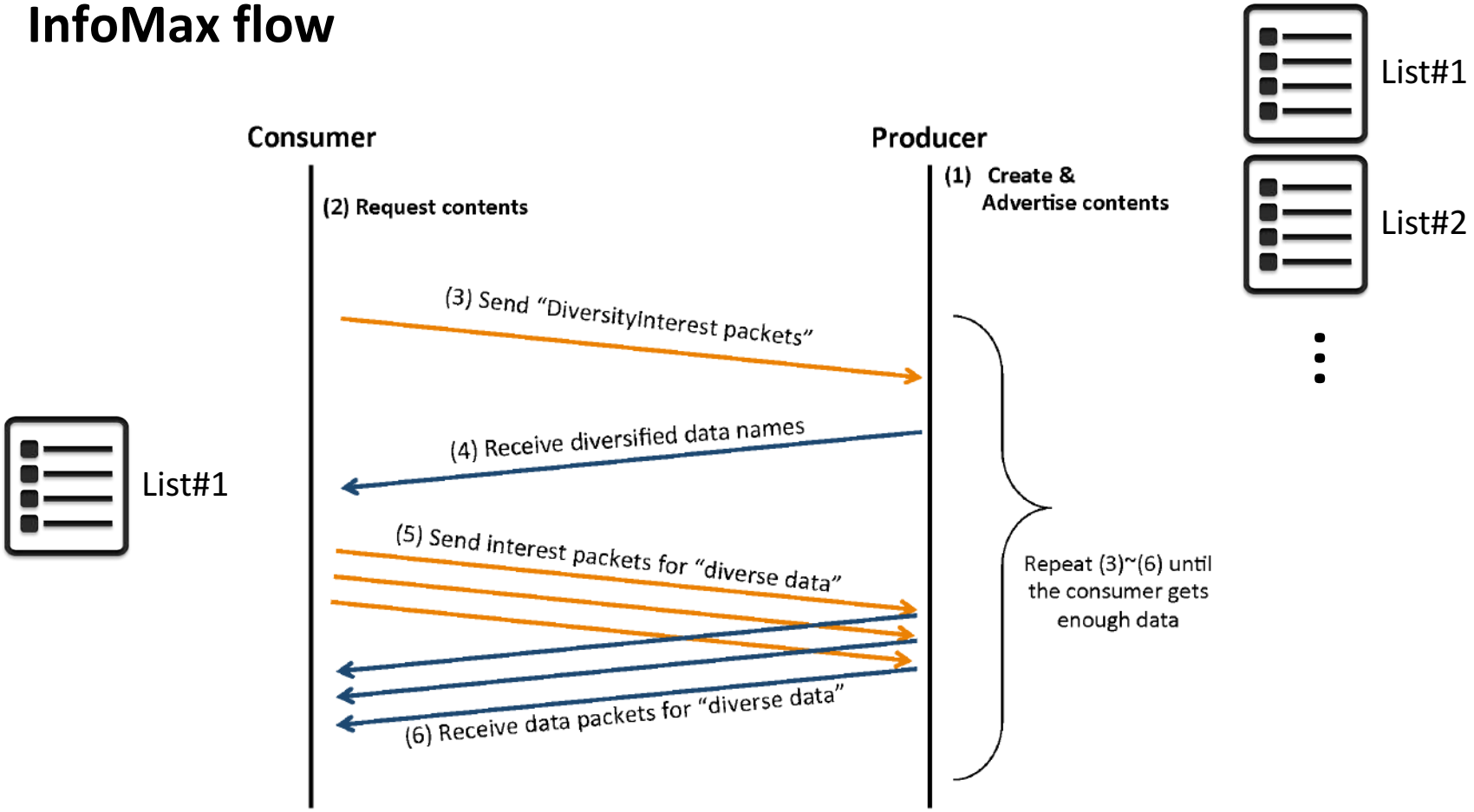
➔ Use sequential lists of ordered data names, which are cacheable

- Allow consumers to control # of retrieved objects to meet individual summarization needs

➔ Consumers can terminate retrieval after they received enough

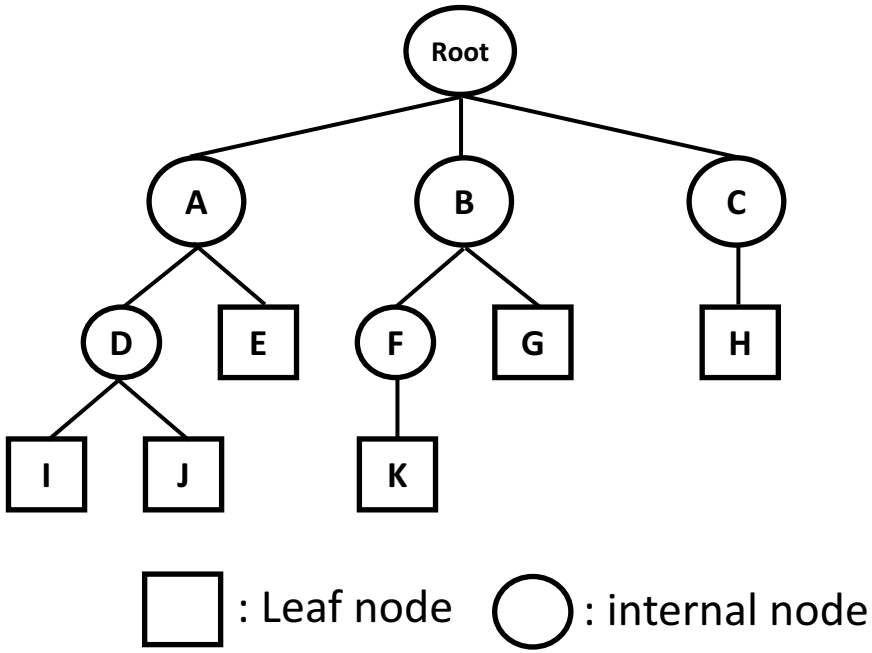
Data Sampling Service

- **InfoMax flow**



Prioritization Algorithm

- **Bottom-up approach**



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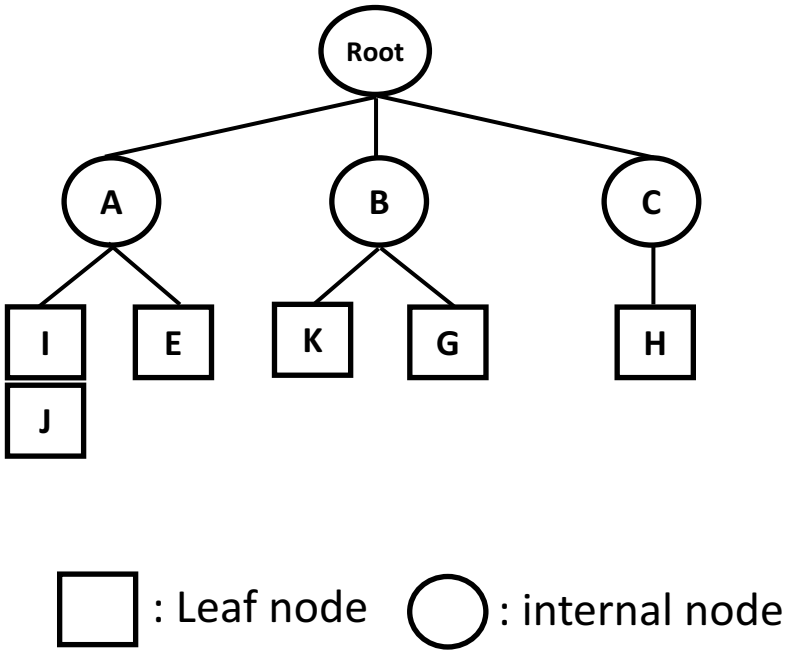

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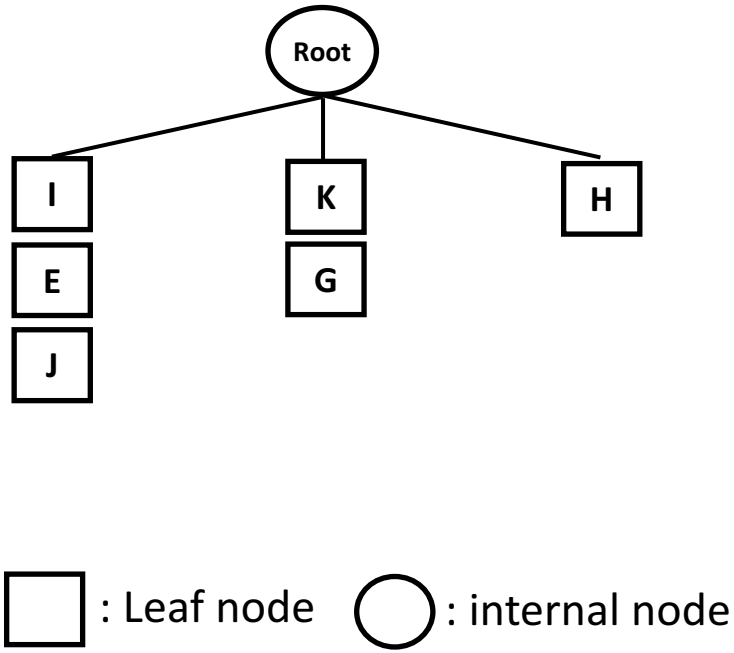

Prioritization Algorithm

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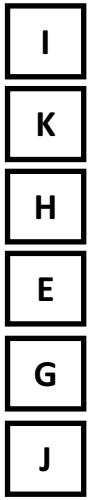
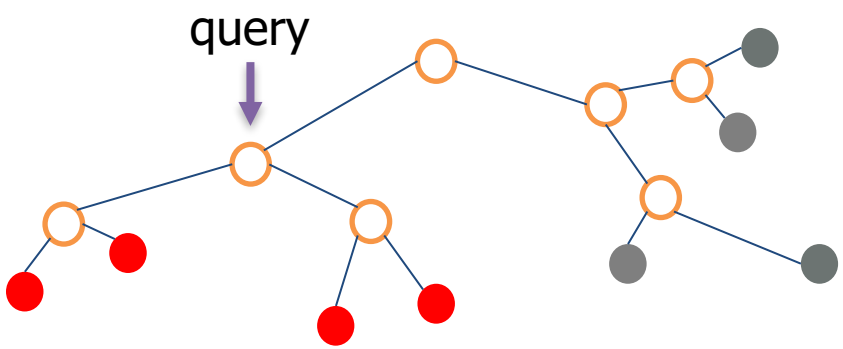


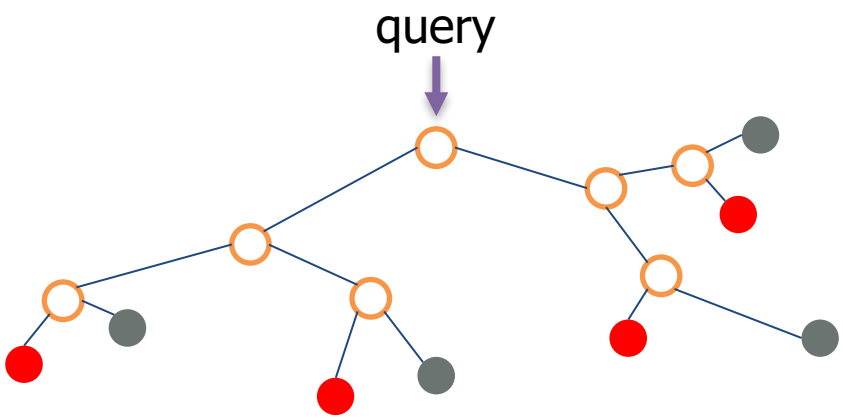
: Leaf node : internal node

Multi-level Summarization

Zoom-in



Zoom-out



Demo

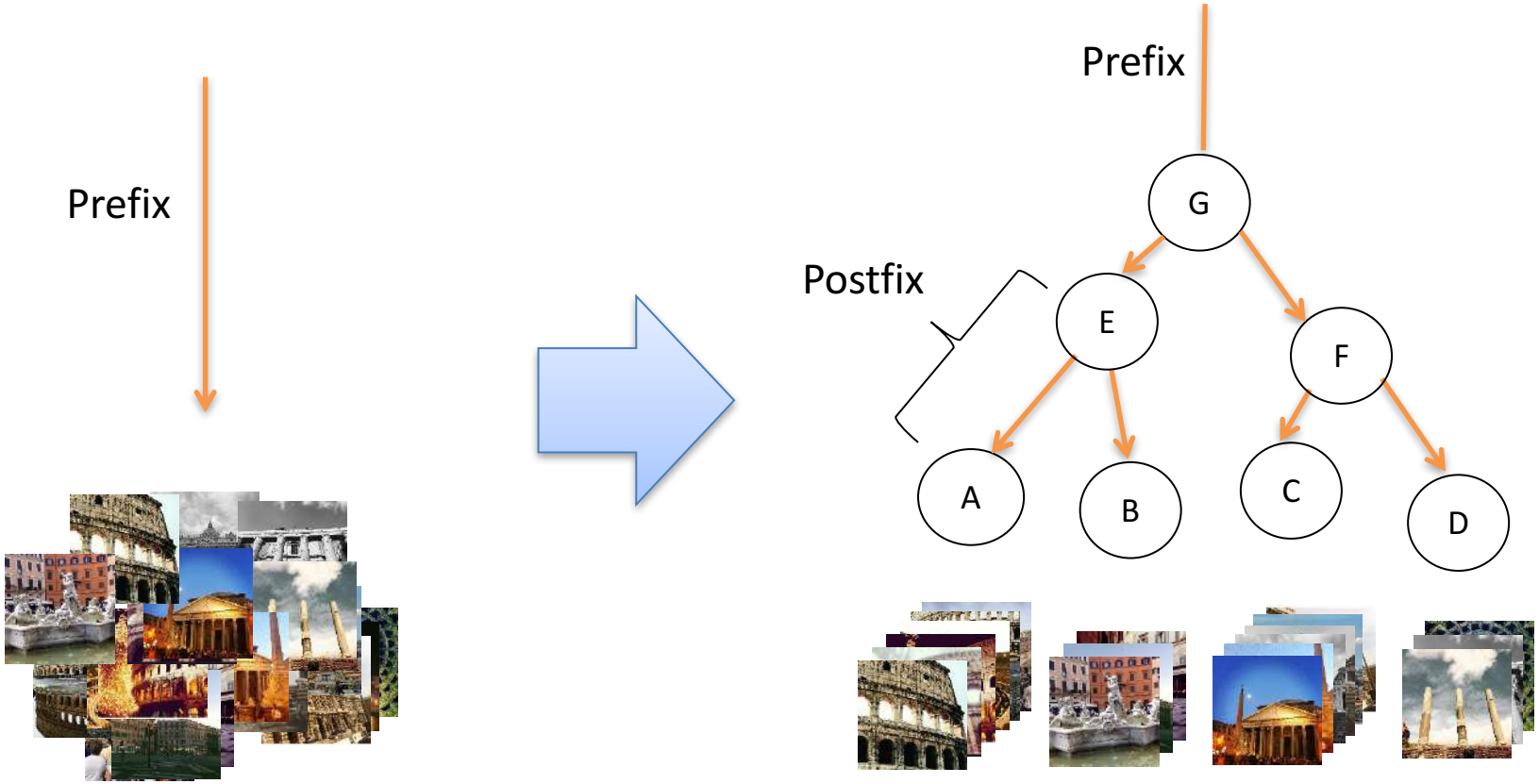
- **Visual Tourism**

Automatic Naming Service

- **Goal: Build a data naming service that allows applications to compose hierarchical names to data in accordance with an application-specific data similarity metric**
  - Automated based on contents similarity
  - Customizable to support various sampling needs

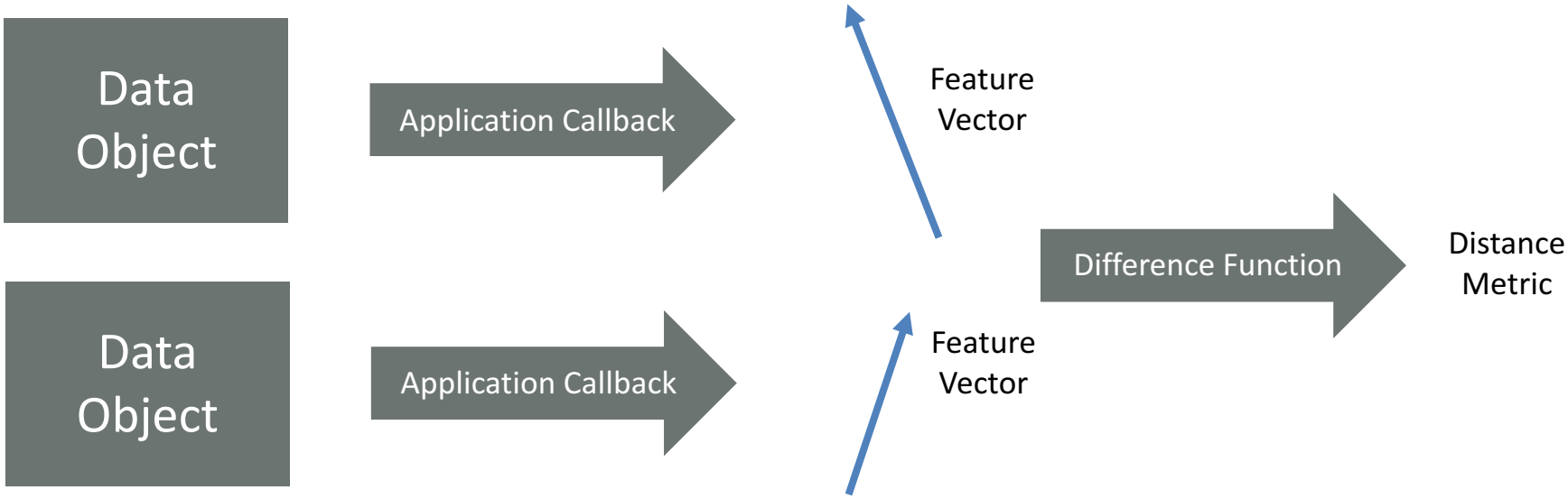
Challenge #1

- **How to automate (postfix) name space design for a given data set such that more similar objects share longer postfix?**



Automatic Naming Service

- **Graph construction**



Automatic Naming Service

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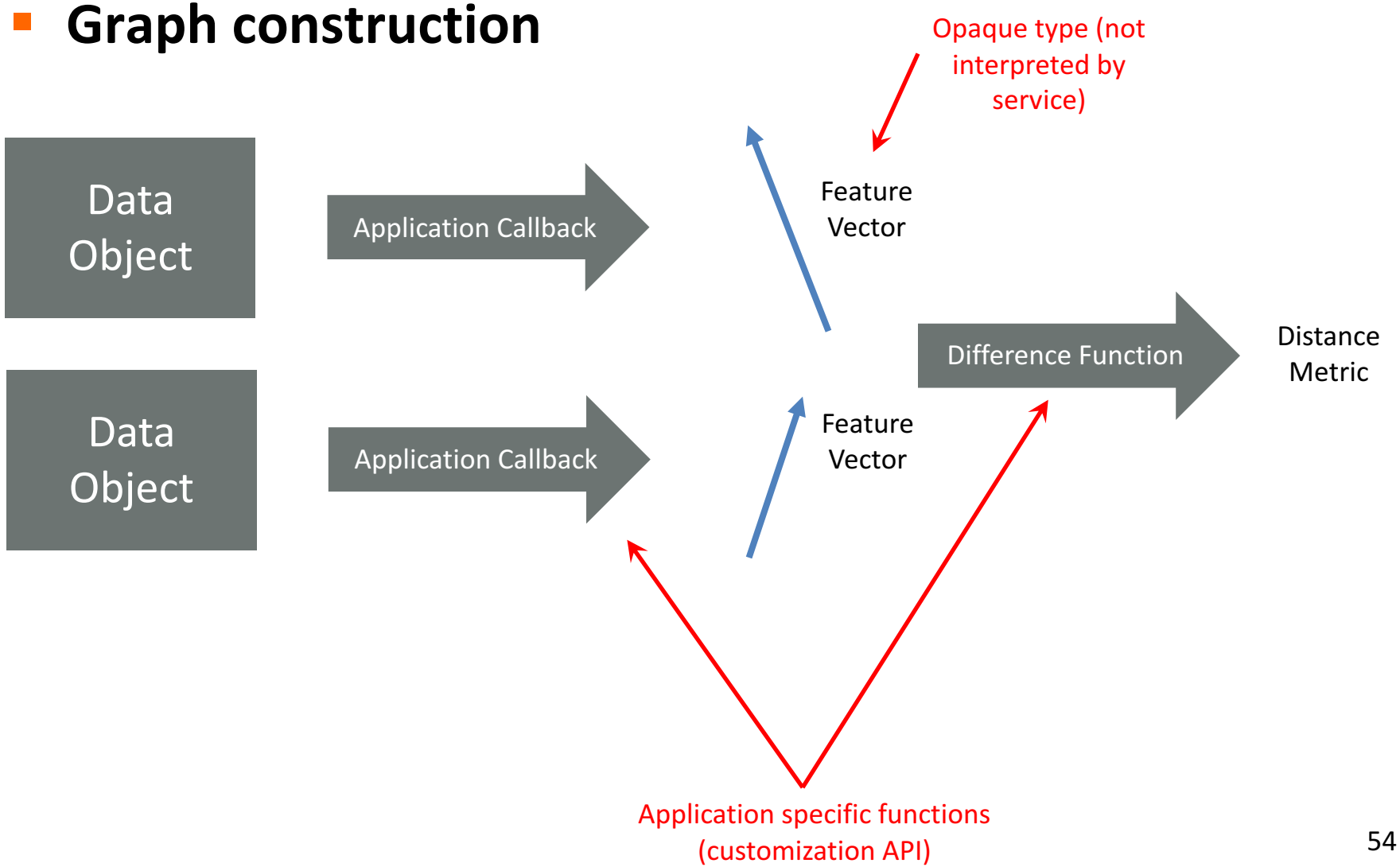
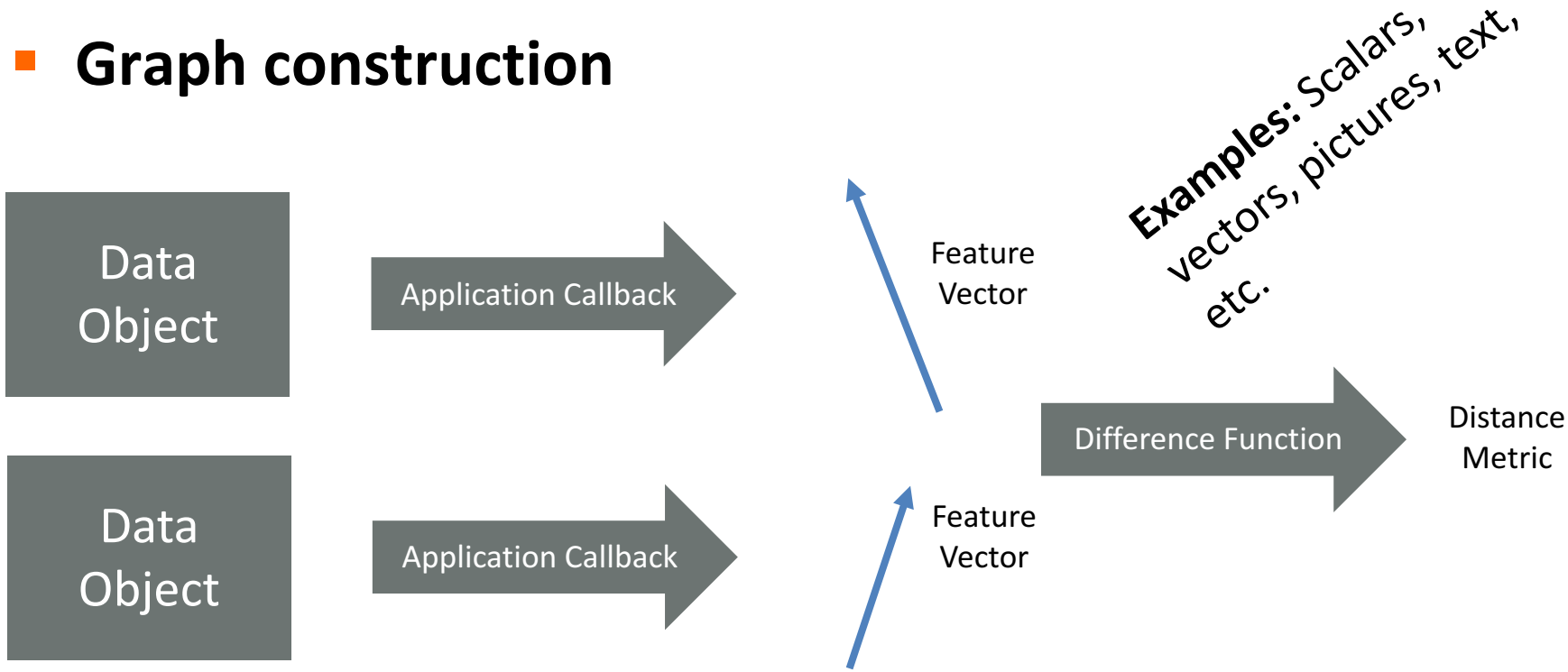


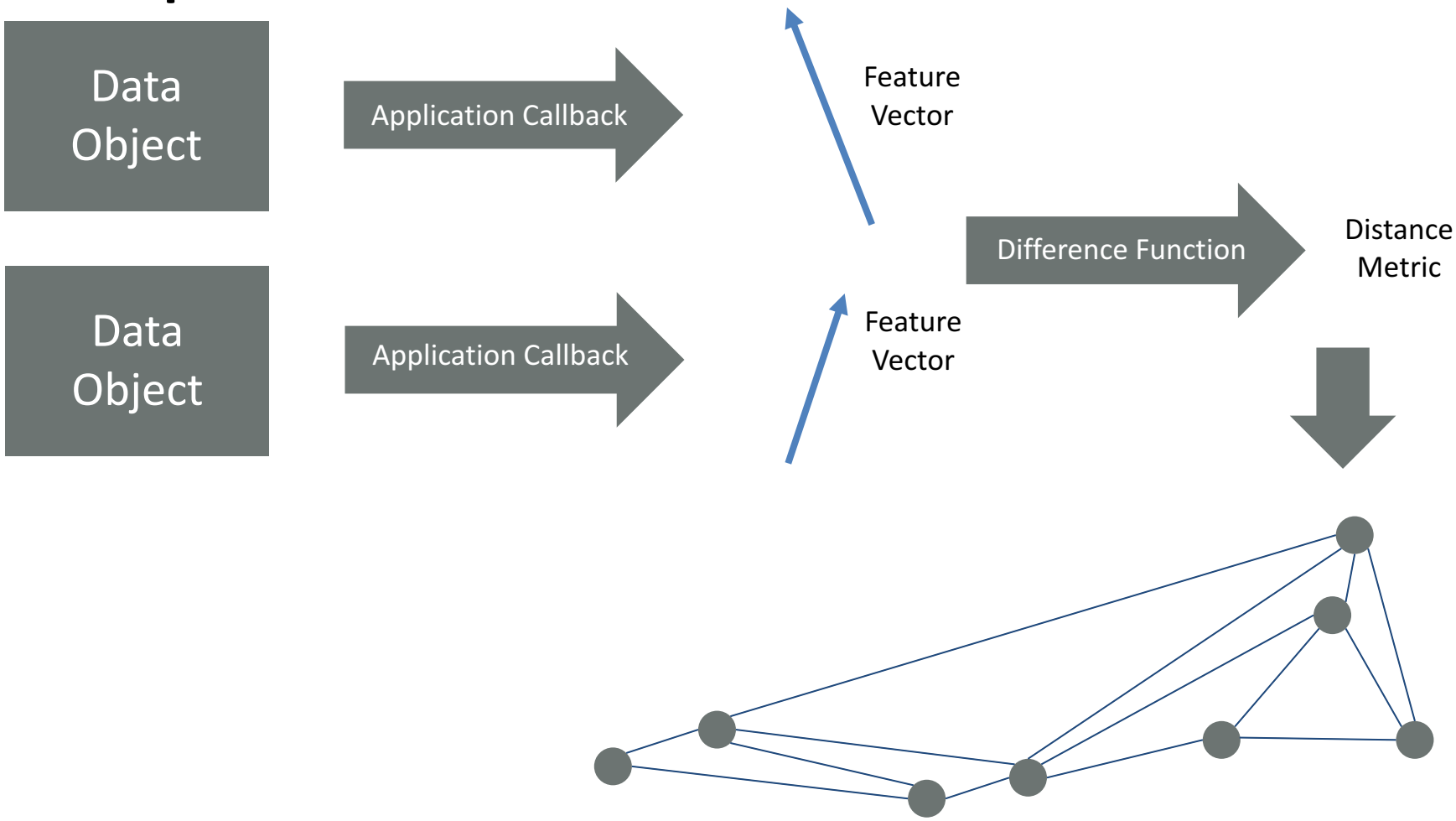
Automatic Naming Service

- **Graph construction**



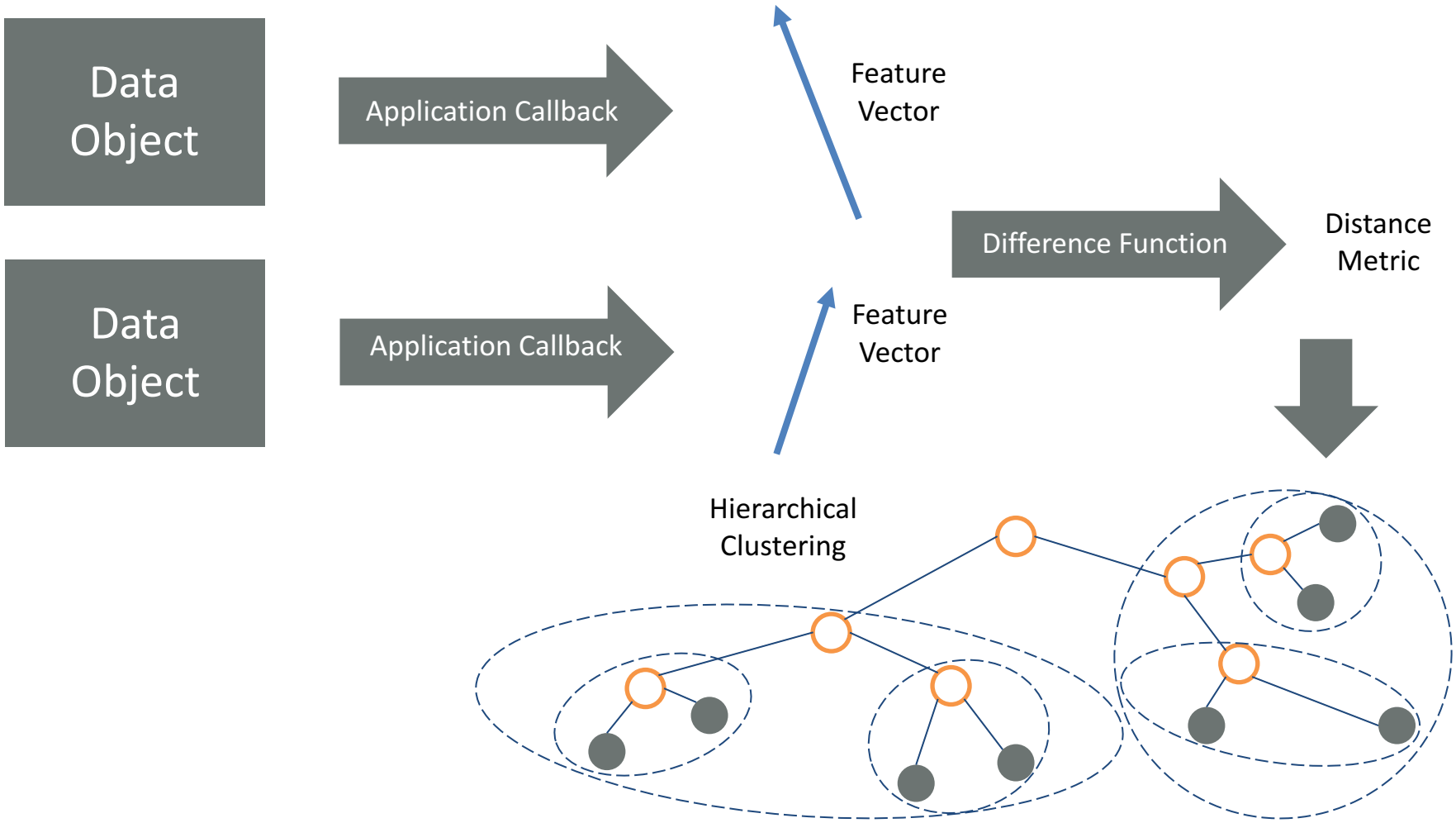
Automatic Naming Service

- **Graph construction**



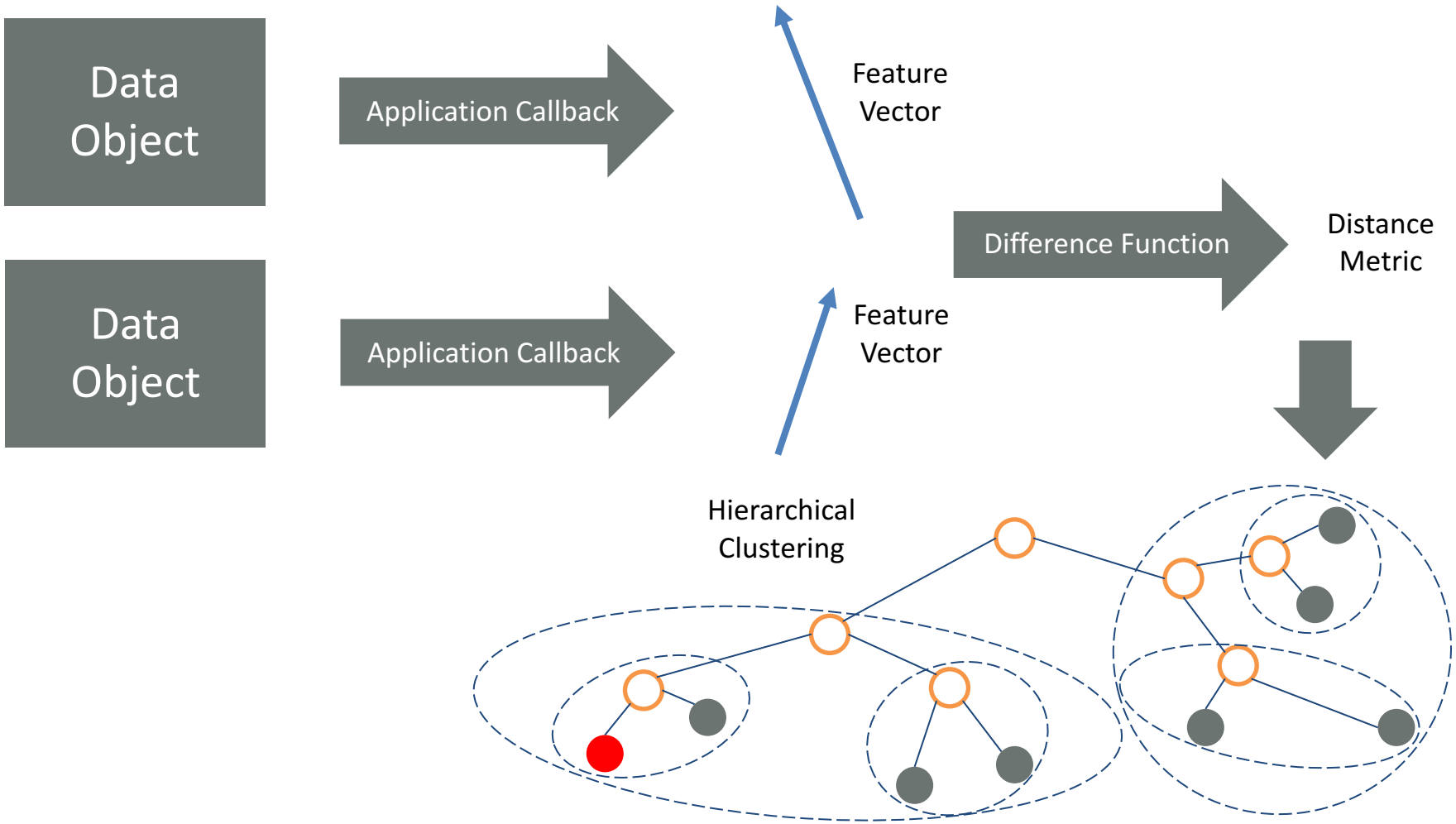
Automatic Naming Service

- **Tree construction**



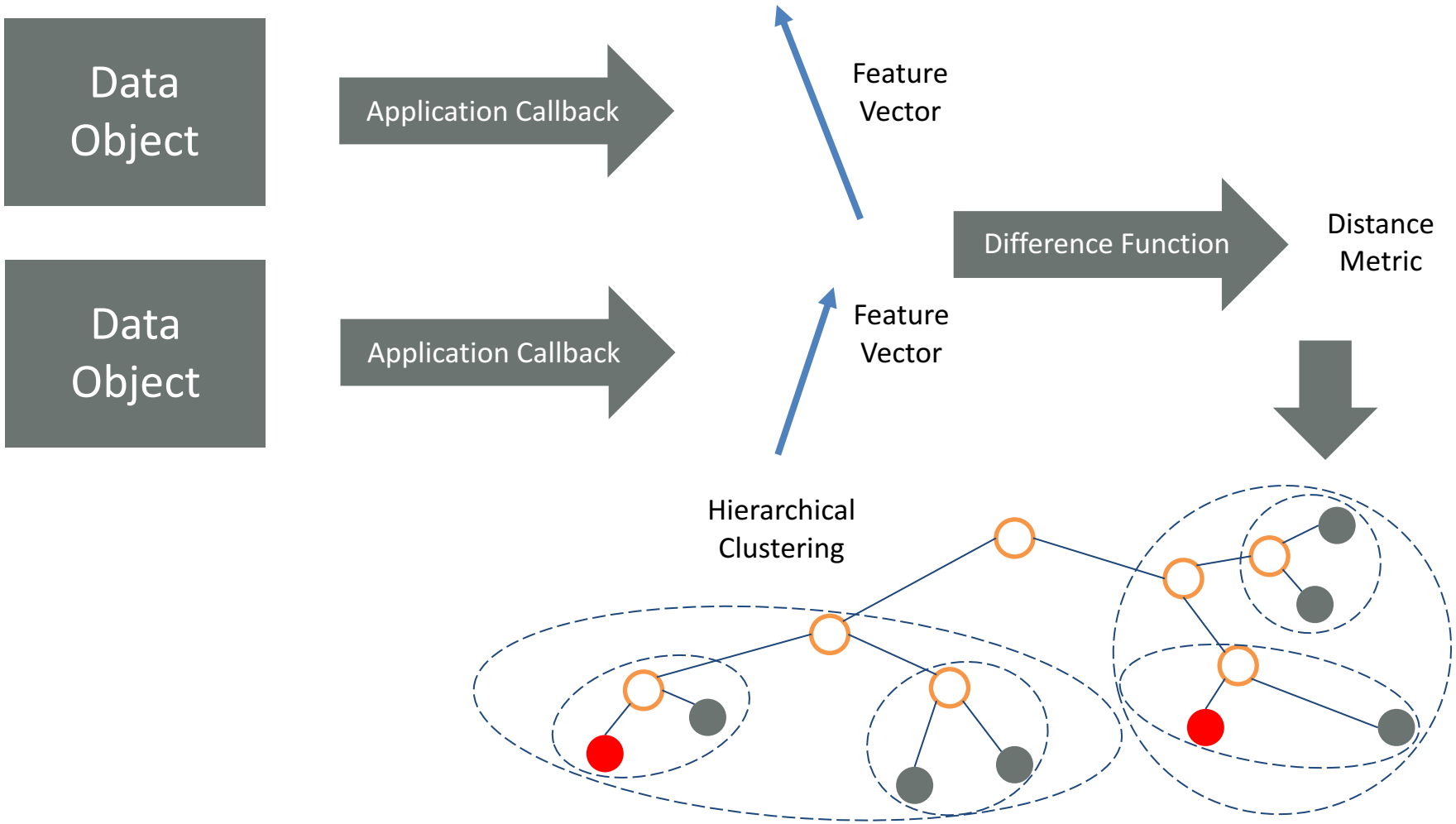
Automatic Naming Service

- **Summarization**



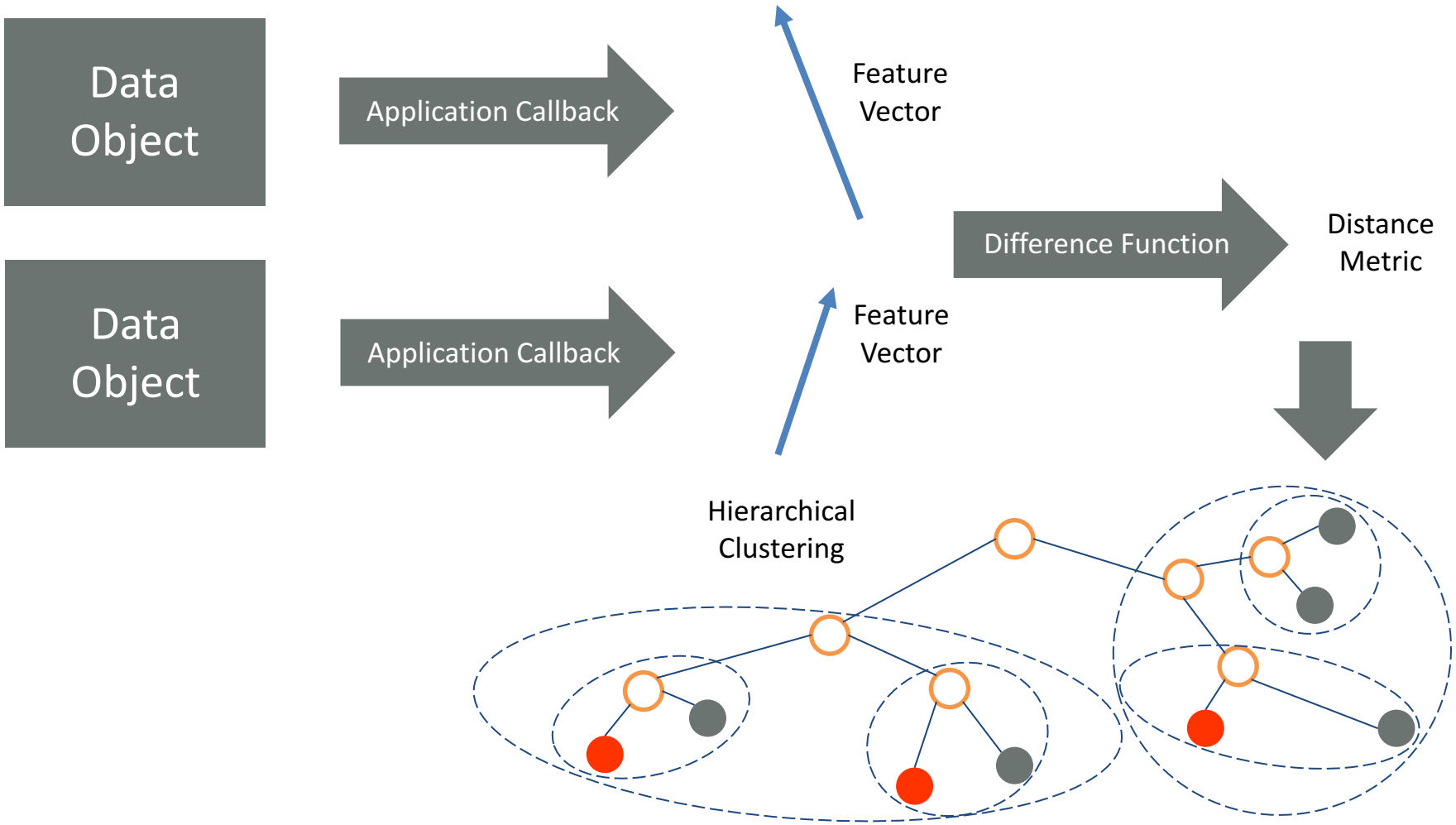
Automatic Naming Service

- **Summarization**



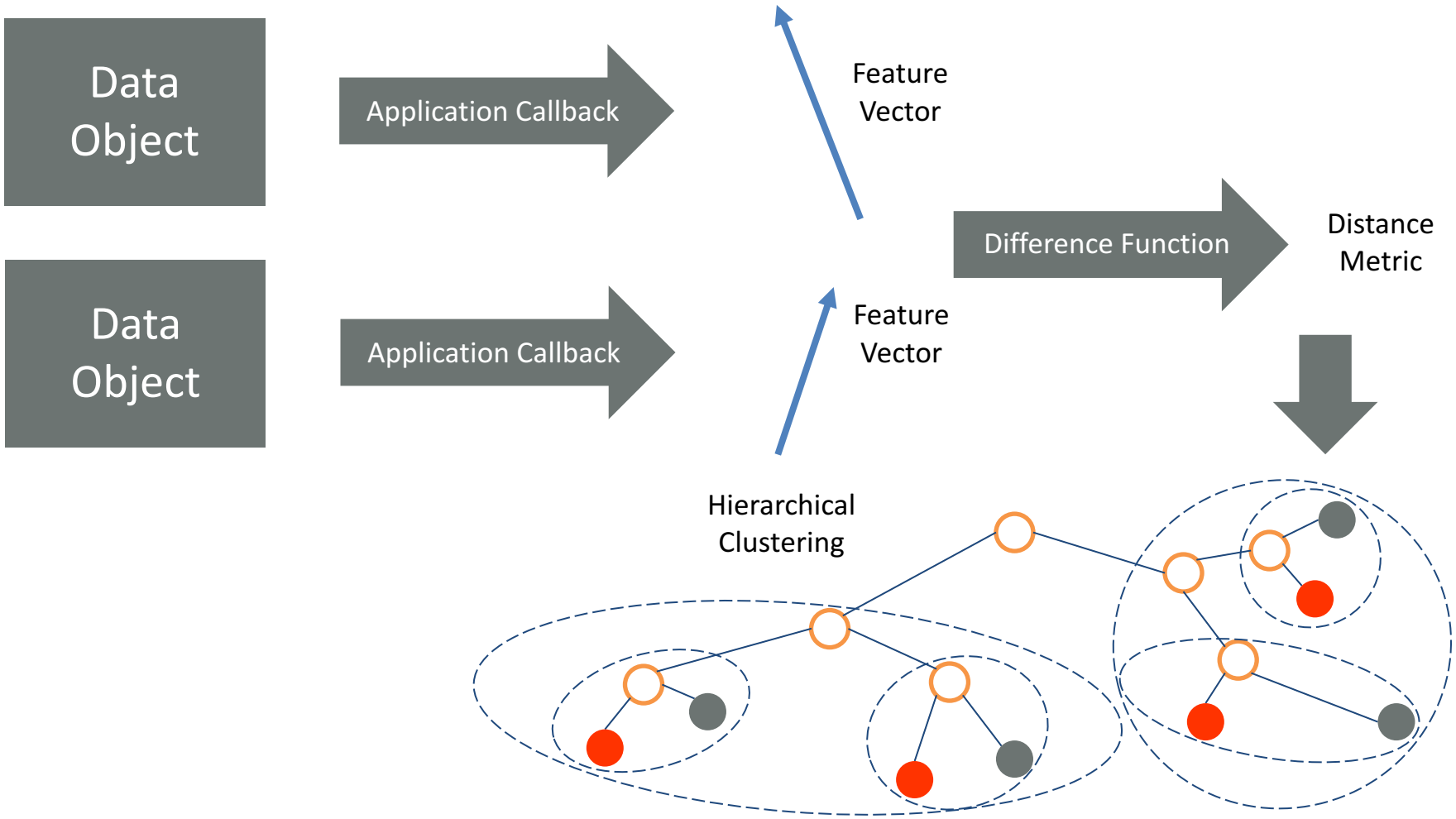
Automatic Naming Service

- **Summarization**



Automatic Naming Service

- **Summarization**



Challenge #2

- **Customizing the InfoMax order**
  - Applications may assign importance weights to data objects
  - How to combine importance weights with InfoMax order?

Custom InfoMax Order

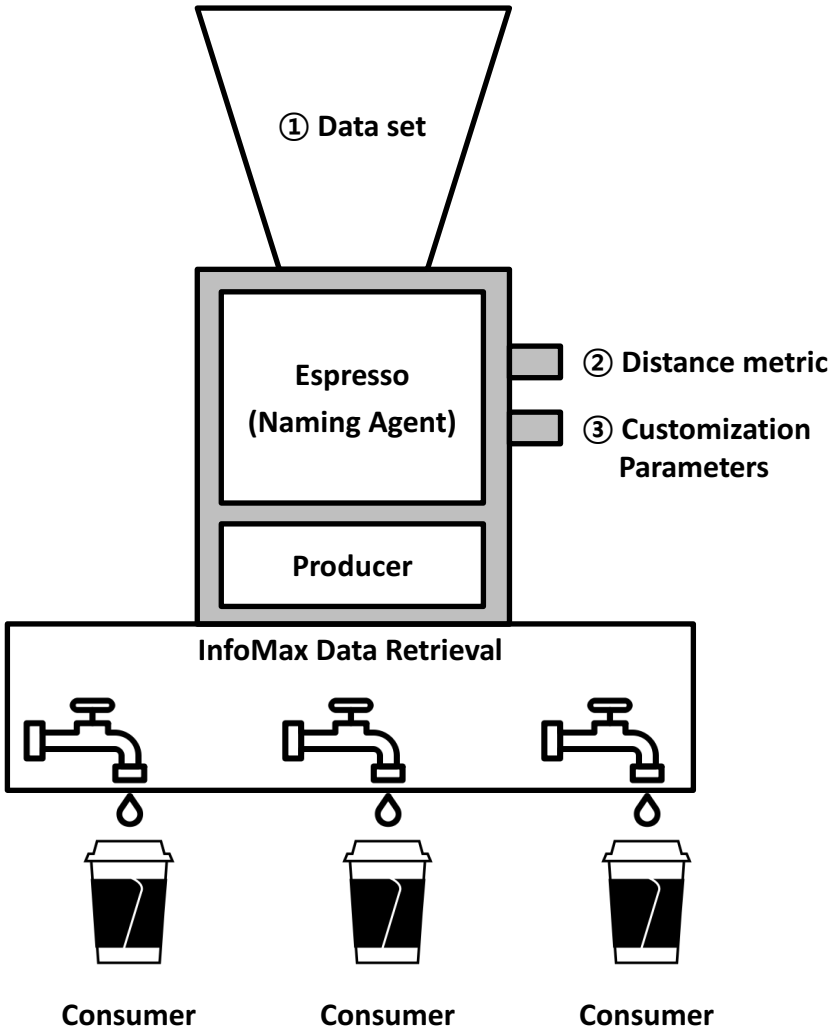
- **3 custom functions from application**
  - **Weight function**
    - » Define weight of leaf node
    - » e.g. size of content, # of likes
  - **Aggregation function**
    - » Define weight of intermediate node
    - » e.g. sum, max, min, average
  - **Branching order**
    - » Define order of traversing branches
    - » e.g. least-visited and left, least-visited and higher aggregate number

Custom InfoMax Order

- **Algorithm variations with 3 dimensions**

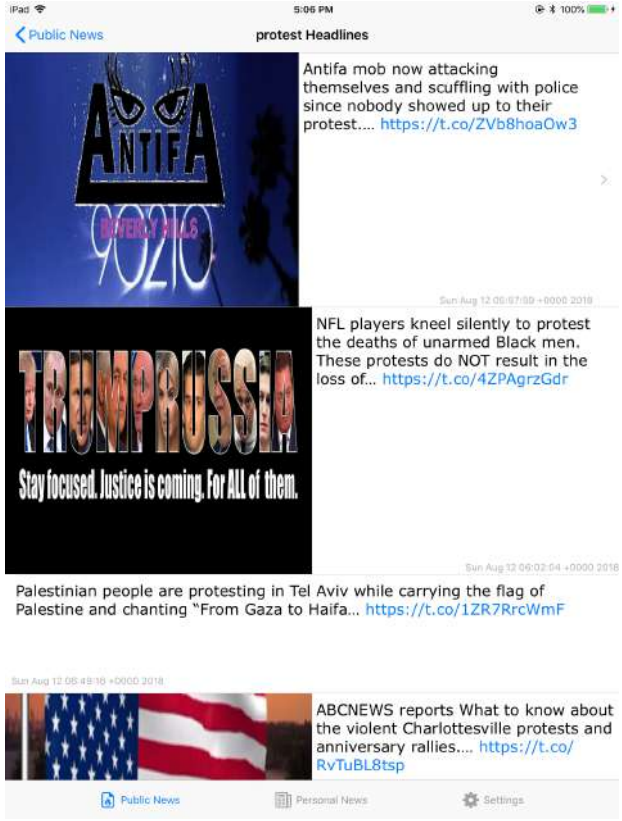
Algo. Name \ Func.		Weight	Aggregatation	Branching order
InfoMax	Diversity only (Default)	*	*	Least visited left first
	Proportional to cluster size	1	Sum	Proportional to sum
	Higher max first	given by application	Max	Least visited Highest first
	Above threshold	give by application	Sum	Least visited Only above threshold

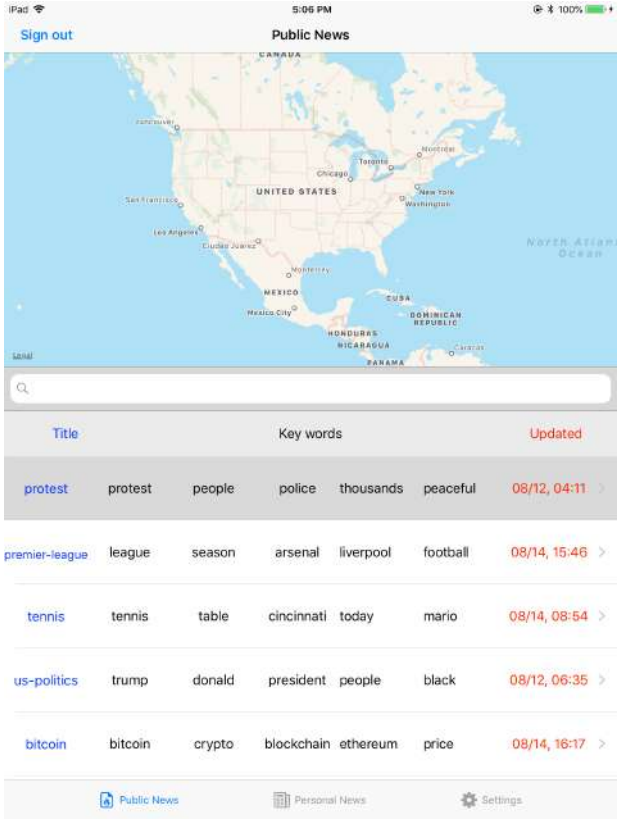
Data Summarization Architecture



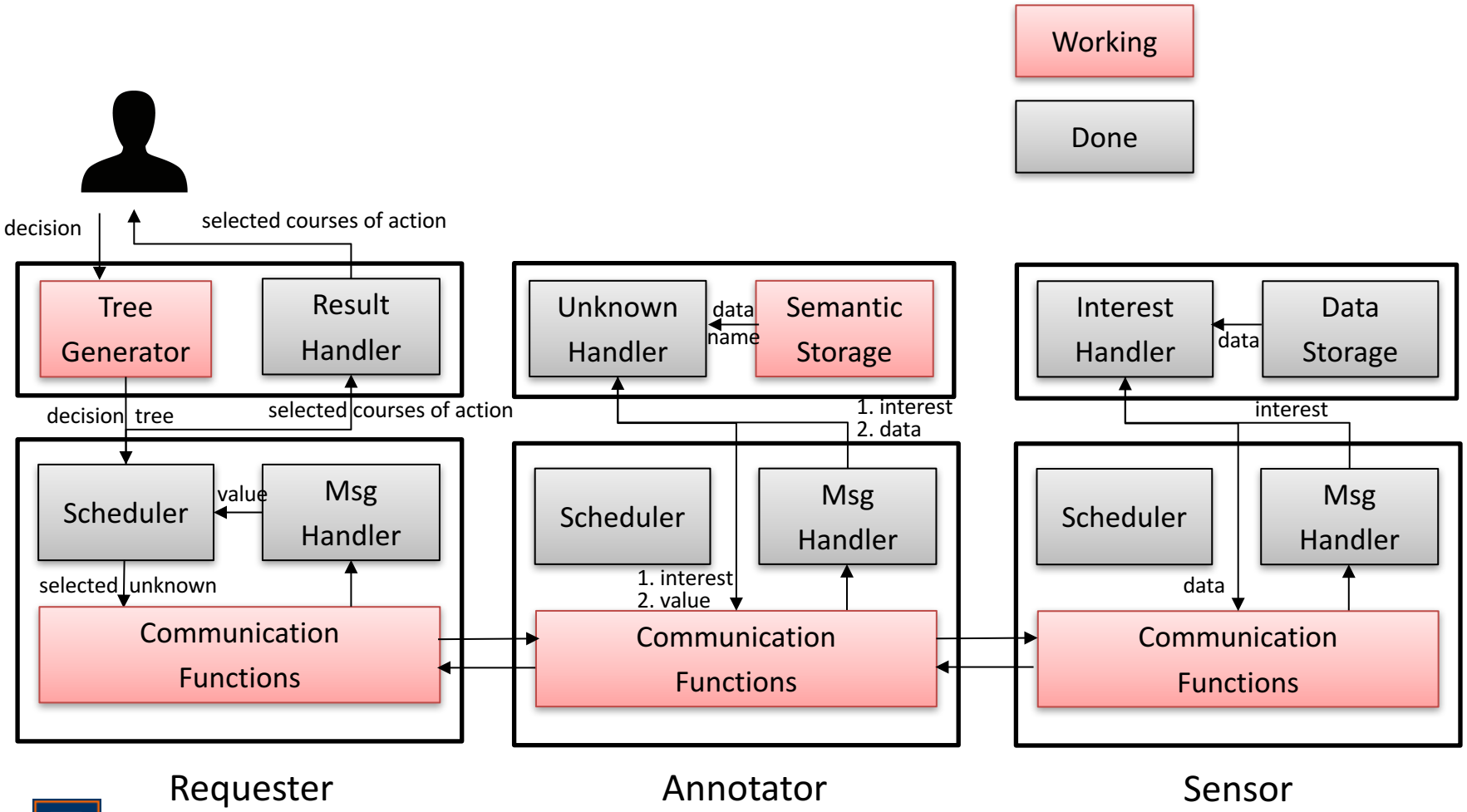
Demo

- **iApollo: Twitter-based News Generator**





Future Work: complete Athena



Conclusions

- **Decision-driven execution** is a new information collection paradigm that directly describes **high level application needs** to network system
- **Athena framework** implements decision-driven execution
- **Two techniques to minimize transmitted data volume:**
  - Prediction-based
  - Redundancy-based
- Remaining task is **to complete Athena framework**